



Measuring the Gap Between Declared and Real Urgency in a Supply Chain Digital Twin

A decision-support layer for the Smart Green Supply Chain programme

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Glossary

Adequacy score	Agreement between declared and computed urgency at a node, scored on $[0, 100]$ with an asymmetric penalty that prices hidden risk at roughly twice false urgency.
AHP	Analytic Hierarchy Process. Elicits criteria weights from pairwise comparisons and validates them through a consistency ratio.
AR(1)	First-order autoregressive process. Used here to extrapolate a node's computed urgency series forward.
AUC	Area under the ROC curve. The probability that a randomly chosen positive case is ranked above a randomly chosen negative one. 0.5 is chance.
Brier skill	A probabilistic forecast's mean squared error, expressed relative to a reference forecaster. Positive beats the reference; negative is worse than it.
Consistency ratio	AHP's internal check on a set of pairwise comparisons. A questionnaire whose ratio exceeds the accepted threshold is refused.
Declared urgency	The urgency an operator states for a node, elicited through the questionnaire. Written U_d .
DIN	Direction de l'Innovation. The entity responsible for ALTEN's own research and development, as distinct from client work.
DISCO	The research programme's serious game and digital twin, simulating a satellite supply chain across 22 component suppliers.
False urgency	The part of the gap where declared urgency exceeds computed urgency. Consumes shared capacity another flow needed.
GLEC Framework	Global Logistics Emissions Council methodology for calculating logistics emissions, aligned with ISO 14083.
Hidden risk	The part of the gap where computed urgency exceeds declared urgency, written $[U_r - U_d]^+$. The dangerous direction, because it raises no alert.
Milestone truth	Whether a milestone was met at the deadline originally committed to. Scripted on the first chain; derived from the indicator trajectory on the second.
Noisy-OR	Probabilistic aggregation in which any block approaching its maximum drives the aggregate toward the maximum, unlike a weighted average.
OEE	Overall Equipment Effectiveness. Composite of availability, performance and quality, used as the productive health proxy of a resource.

Physical Internet	Logistics paradigm in which shipments move through shared, interoperable networks rather than dedicated ones.
PR-AUC	Area under the precision-recall curve. More informative than AUC when positives are rare.
Precision@k	Among the k nodes a ranking selects, the share that actually fail. The form the question takes when a coordinator can only look at three suppliers.
Real urgency	The urgency computed at a node from six indicator blocks, before any human states anything. Written U_r .
Rank / tier	A supplier's depth in the network, counted from the final customer.
Ripple effect	The amplification of a local supply disruption as it propagates through downstream tiers.
Turn	One week of simulated time. A campaign turn closes when every node has submitted its weekly review.

Résumé

Dans une chaîne d'approvisionnement, l'urgence qu'un opérateur déclare diverge souvent de celle que l'on peut calculer à partir de l'état opérationnel réel. Sur-déclarer gaspille une capacité partagée ; sous-déclarer laisse un risque de rupture invisible. Aucun jumeau numérique existant, publié ou breveté, ne mesure cet écart.

Cette mission, menée à la Direction de l'Innovation d'ALTEN dans le programme Smart Green Supply Chain, a conçu, développé et éprouvé une couche logicielle qui le mesure. L'urgence réelle est agrégée à chaque nœud depuis six blocs d'indicateurs par un OU probabiliste pondéré, puis propagée sur le réseau ; l'urgence déclarée est élicitée par un questionnaire de comparaisons par paires contrôlé en cohérence ; leur écart est noté sur $[0, 100]$ par une pénalité asymétrique. Le système livré, SupplyScore, compte 73 modules Python couverts par 1 831 fonctions de test.

Il répond à la question que pose un coordinateur : quel fournisseur regarder cette semaine. Sur les six tours qui précèdent la mise au jour d'une crise, l'écart désigne trois nœuds, tous trois futurs défaillants, sans une seule fausse alerte ; le plus mal noté est signalé huit tours avant de rater son jalon, alors que son propre responsable déclare une urgence de 0,001.

La moitié la plus instructive de la mission a consisté à tenter de détruire ce résultat, par trois épreuves. La couche de prévision a d'abord échoué de façon diagnosticable, un seul bloc l'expliquant, et le correctif évident s'est révélé faux à la mesure. Une seconde chaîne, conçue hostile à la première, montre que l'écart n'y désigne rien : les trois fournisseurs défaillants déclarent les urgences les plus fortes tout en écrivant qu'ils différeront l'alerte à leur client, si bien que l'instrument détecte celui qui se méprend et non celui qui se tait. Enfin, un témoin d'une ligne, ne lisant qu'une échéance et un délai nominal, classe les jalons ratés mieux que cinq cents trajectoires de Monte-Carlo à six horizons sur huit. Ce qui borne ces chiffres est l'échantillon et non la construction ; deux mois de mission restent à courir pour l'élargir.

Abstract

In a supply chain, the urgency an operator declares often diverges from the urgency computable from the real operational state. Over-declaring wastes shared capacity; under-declaring leaves rupture risk invisible. No existing digital twin, published or patented, measures that gap.

This mission, carried out at the Direction de l'Innovation of ALTEN within the Smart Green Supply Chain programme, designed, built and stress-tested a software layer that measures it. Real urgency is aggregated at each node from six indicator blocks through a weighted probabilistic OR and propagated across the network; declared urgency is elicited through a consistency-gated pairwise questionnaire; their gap is scored on $[0, 100]$ with an asymmetric penalty. The delivered system, SupplyScore, comprises 73 Python modules covered by 1 831 automated test functions.

It answers the question a coordinator actually asks: which supplier to look at this week. Over the six turns before a crisis becomes public, the gap flags three nodes, all three of which later miss a delivery commitment, with no false alarm on any turn; the worst-scoring of them is flagged eight turns before it misses, while its own manager declares an urgency of 0.001.

The more instructive half of the mission was spent trying to destroy that result, through three tests. The forecast layer failed first and failed diagnostically, one block explaining it, and the obvious repair proved wrong under measurement. A second chain, specified to be hostile to the first result, shows the gap selecting nothing there: the three failing suppliers declare the highest urgencies in the chain while writing that they will delay warning their customer, so the instrument detects a declarant who misjudges rather than one who stays silent. Finally, a one-line baseline reading only a due date and a nominal lead time ranks the missed milestones better than five hundred Monte Carlo trajectories at six horizons of eight. What bounds these figures is the sample rather than the construction, and two months of the mission remain in which to widen it.

Resumen

En una cadena de suministro, la urgencia que declara un operador difiere a menudo de la urgencia calculable a partir del estado operativo real. Declarar de más desperdicia capacidad compartida; declarar de menos deja invisible el riesgo de ruptura. Ningún gemelo digital existente, publicado o patentado, mide esa diferencia.

Esta misión, realizada en la Dirección de Innovación de ALTEN dentro del programa Smart Green Supply Chain, diseñó, desarrolló y sometió a prueba una capa de software que la mide. La urgencia real se agrega en cada nodo a partir de seis bloques de indicadores mediante un O probabilístico ponderado y se propaga por la red; la urgencia declarada se obtiene mediante un cuestionario de comparaciones por pares con control de coherencia; su diferencia se puntúa en $[0, 100]$ con una penalización asimétrica. El sistema entregado, SupplyScore, consta de 73 módulos Python cubiertos por 1831 funciones de prueba automatizadas.

Responde a la pregunta que se hace un coordinador: qué proveedor mirar esta semana. En los seis turnos previos a que una crisis se haga pública, la diferencia señala tres nodos, los tres futuros incumplidores, sin una sola falsa alarma; el peor puntuado se señala ocho turnos antes de incumplir su hito, mientras su propio responsable declara una urgencia de 0,001.

La mitad más instructiva de la misión consistió en intentar destruir ese resultado, mediante tres pruebas. La capa de predicción falló primero y de forma diagnosticable, con un solo bloque como causa, y la corrección evidente resultó errónea al medirla. Una segunda cadena, diseñada hostil a la primera, muestra que allí la diferencia no señala nada: los tres proveedores que incumplen declaran las urgencias más altas de la cadena mientras escriben que retrasarán avisar a su cliente, de modo que el instrumento detecta a quien se equivoca y no a quien calla. Por último, un testigo de una línea, que solo lee una fecha de vencimiento y un plazo nominal, clasifica los hitos incumplidos mejor que quinientas trayectorias de Monte Carlo en seis horizontes de ocho. Lo que acota estas cifras es la muestra y no el diseño, y quedan dos meses de misión para ampliarla.

Preamble

Status of the results. This report is written at the end of the fourth month of a six-month mission. Two months of work remain, and one strand of it is deliberately still open: the instrument is being run against additional supply chains to widen the evidence base behind every rate quoted here. The construction, the measurements and the diagnoses are stable and reproducible from the artefacts listed in Section 4.3. The numbers attached to them are those of the campaigns completed to date and may move as that base widens. Where a figure rests on a small sample, the report says so at the point of use rather than in a caveat at the end.

Confidentiality. The project is designated confidential by ALTEN in the annex to the placement agreement. This report carries a confidentiality marking on every page. It contains internal scoring parameters, the architecture of an unreleased software product and results not published elsewhere. It is submitted to ESTIA for examination only.

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1 Introduction

A supply chain coordinator opens the week with a list of nodes and a finite amount of attention. Somewhere on that list is the supplier that will cause next month's line stoppage. Nothing on the screen says which one it is.

That is the practical shape of the problem this mission addressed. Modern supply chains are monitored through indicators that each measure one dimension at a time, while the decision the coordinator actually faces is multi-dimensional: cost against deadline against supplier reliability, carbon, and more. When the tooling cannot combine those dimensions, the coordinator combines them by judgment. Judgment is then the real decision layer of the system, and it is the layer nobody measures.

The consequences are not evenly distributed. Declaring more urgency than a situation warrants consumes shared transport and production capacity that another flow needed. Declaring less leaves a risk that nobody is watching, and by the time it becomes visible the window to act on it has closed. The second error is the more expensive, and it is the harder one to detect, precisely because an under-declared risk generates no direct signal.

This mission was carried out at the Direction de l'Innovation of ALTEN, within the Smart Green Supply Chain Digital Twin research programme, between 13 April and 16 October 2026. Its subject is the layer that sits before any planning decision: the urgency signal itself. The question driving the work was whether the gap between what an operator declares and what the operational state actually supports can be measured well enough, inside a working digital twin, to be acted upon.

Building that measurement was the first half of the mission. Trying to destroy it was the second, and it is the half that produced the transferable results. Three tests were designed against the system rather than for it. Each was run to completion and each returned a usable answer.

The gap is measurable and it carries information. On the first campaign, the nodes the system flags in the six turns before a crisis becomes public are, without a single false alarm, nodes that go on to miss a delivery commitment, and the worst-scoring of them is flagged eight turns before it misses while its own manager declares an urgency of 0.001. Turning that flag into a calibrated probability attached to a named week is a further step, and it failed at the first attempt. It failed diagnostically: one component was responsible, the mechanism was identified, a repair was specified, carried out and re-measured on the same frozen forecasts, and the repair the diagnosis originally recommended turned out under measurement to be the wrong one.

A second supply chain was then built, deliberately hostile to the first result: five times the nodes, no exogenous shock at all, and milestone truth computed from the indicators rather than written by the scenario author. There the gap selects nothing, and the reason draws a boundary the specification never anticipated. The suppliers that fail declare the highest urgencies in the chain from the opening turn while writing, in the same forms, that they intend to delay telling their customer. The instrument detects a declarant who misjudges their own state; it cannot detect one who judges it correctly and manages what they say.

The third test was aimed at the system's most expensive component. A baseline of one line, reading only a due date and a nominal lead time, was scored against the five hundred Monte

Carlo trajectories the forecast layer runs per node per turn. It ranks the failures better at six horizons of eight. What the simulation contributes is calibration and long-horizon ordering, not the short-horizon discrimination its interface implies, and the line of the model responsible is named.

Alongside these, one measurement was not sought and is the most operationally uncomfortable: displaying the system's own forecast to the person about to declare changes what they declare, in both directions and at both scales tested.

The report is organised as the mission was run. Section 2 presents ALTEN, the Direction de l'Innovation and the research programme this work sits inside, including the company's corporate social responsibility policy and how the mission relates to it. Section 3 states the problem that justified opening the subject. Section 4 gives the assignment, its objectives, its indicators and its deliverables. Section 5 describes how the project was run: who was involved, what resources were available, which working method was applied, how the six months were phased, and which risks were carried. Section 6 reviews the existing methods, published and patented, and identifies what none of them provides. Section 7 reports the work carried out, the three tests above and what they returned, including the problems met along the way and how they were resolved. Section 8 assesses the results against the objectives, sets out what remains to be done, and closes with a personal assessment against the ESTIA engineering competency framework.

2 The Company

2.1 The ALTEN Group

ALTEN is a French engineering and technology consulting group founded in 1988, with its head office in Boulogne-Billancourt. It sells engineering expertise rather than products: its consultants are placed on the technical projects of client companies, across the aerospace, defence, energy, transport, telecommunications, health and finance sectors. The group operates in more than thirty countries, and France remains its largest market.

Two features of that model matter for understanding this mission. First, ALTEN's asset is the technical capability of its people, so anything that raises the level of a whole practice has leverage that a single client project does not. Second, a consulting group sees the same problem recur across many clients, which puts it in an unusual position to notice that a difficulty is structural rather than particular. The urgency assessment problem treated in this report is of that kind.

2.2 The Direction de l'Innovation

The Direction Innovation Numérique (DIN) is the entity responsible for ALTEN's own research and development, as distinct from the work its consultants deliver to clients. It operates a network of ALTEN Labs in France and abroad, and brings together permanent researchers, doctoral candidates, consultants, interns and apprentices on multidisciplinary projects. Its output is not billable hours but reusable assets: methods, prototypes and expertise that consultants and clients can subsequently draw on.

This mission took place at the Toulouse Lab, within the Smart Green Supply Chain research programme. The programme pursues three aims simultaneously. The lean aim targets waste, meaning cost, delay and excess stock. The green aim targets the carbon footprint of each transport leg, working to the Global Logistics Emissions Council (GLEC) Framework v3.2 (Smart Freight Centre, 2023) and ISO 14083:2023 (International Organization for Standardization, 2023). The predictive aim, where this work sits, targets the anticipation of disruption.

The programme's central artefact is DISCO, a serious game and digital twin simulating a complete supply chain for a satellite product across 22 component suppliers (Baxas, 2025). Successive placements have built its components: the Physical Internet routing module (Baulain et al., 2024), the supplier risk scorer (Kabouche, 2025), the per-route carbon quantification (Cordova, 2024) and the transport management interface (Mondelice, 2025). The work described here adds a layer none of them covers, and it ran alongside a parallel placement on the network representation and geographic modelling side of the same programme, discussed in Section 5.1.

Figure 1 shows the supplier network players work with and the three dimensions their decisions are scored on. It is the environment the present work was designed to be compatible with, although the delivered system runs alongside DISCO rather than inside it, for reasons given in Section 4.2.

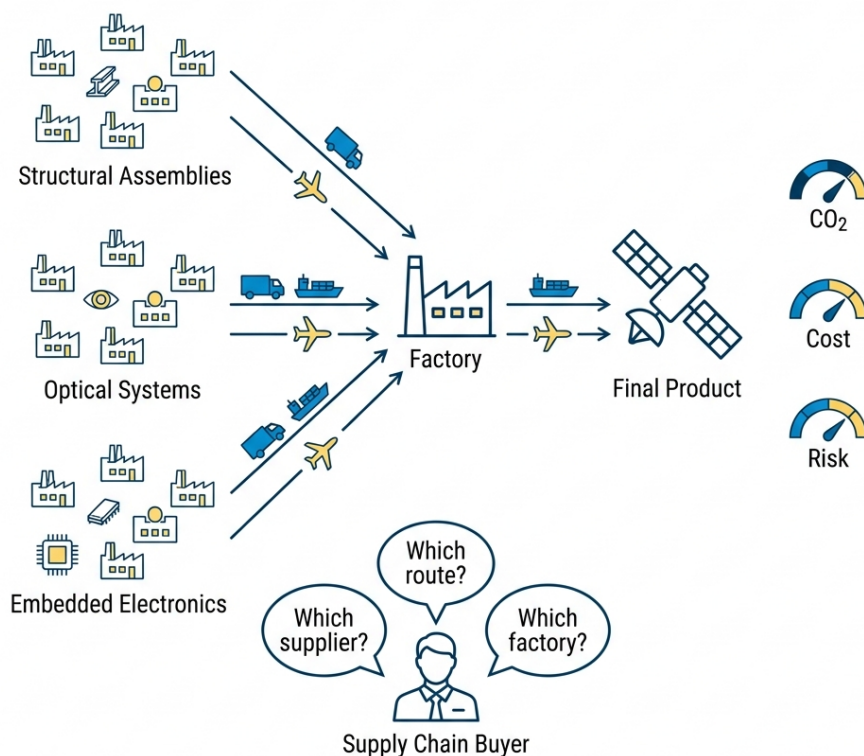


Figure 1: The DISCO serious game network

2.3 Politique RSE d'ALTEN

ALTEN a adhéré au Pacte Mondial des Nations Unies en 2010 et structure depuis sa politique de responsabilité sociétale autour de trois piliers : l'humain, l'environnement et l'innovation durable. La démarche se mesure autant qu'elle se déclare. En 2025, le Groupe a obtenu un score EcoVadis de 85/100, niveau Platinum, qui le place dans le premier centile mondial, et a confirmé sa note A- au Carbon Disclosure Project. Sa trajectoire de réduction des émissions a été validée par la Science Based Targets initiative en 2023, avec un objectif de neutralité carbone à l'horizon 2050.

Sur le pilier humain, le Groupe réunit plus de cent nationalités et a signé les *Women's Empowerment Principles* des Nations Unies. L'index d'égalité professionnelle publié au titre de 2025 s'élève à 88/100 pour ALTEN Sud-Ouest, l'entité juridique dans laquelle s'est déroulée cette mission, et le programme INSPIRE 2026-2030 en porte la suite.

Sur le pilier environnemental, les actions internes sont concrètes et vérifiables : 100 % d'électricité renouvelable pour les bâtiments français dont la puissance souscrite dépasse 36 kVA, 80 % des surfaces couvertes par le tri sélectif, et une certification ISO 14001 en cours d'extension. La sensibilisation passe par des ateliers Fresque du Climat et des formations aux éco-gestes.

C'est le troisième pilier qui éclaire le mieux cette mission. ALTEN consacre 31 % de sa R&D à l'innovation environnementale, et le programme dans lequel ce stage s'est inscrit en fait partie : un programme *Smart Green Supply Chain*, où les émissions de CO₂ constituent l'un des six blocs du modèle d'urgence développé ici, au même rang que le coût ou le

délai, et non une correction appliquée après coup. La politique RSE du Groupe n'a donc pas seulement encadré ce travail : elle en a déterminé une variable de modélisation.

Prise de recul. Cette place de premier rang accordée au carbone a produit un enseignement que je n'attendais pas. Lors de la première campagne expérimentale, le bloc carbone s'est révélé le moins sollicité des six, non par choix de modélisation mais faute de données pour l'alimenter : les indicateurs d'émissions par trajet existent au niveau du programme, pas encore au niveau du nœud et de la semaine, qui est la maille où l'urgence se calcule. Une ambition environnementale peut ainsi être pleinement inscrite dans un modèle et rester la moins active de ses dimensions, parce que la mesure ne descend pas encore à la maille de la décision. L'écart ne tient pas à l'intention mais à la granularité disponible, et le comblement des indicateurs mené en Section 7.5 l'a réduit sans le fermer. C'est, à mon sens, la contribution RSE la plus utile de cette mission : avoir rendu visible et chiffrable le point exact où la donnée environnementale manque pour peser sur un arbitrage opérationnel.

3 Problem Statement

3.1 Why ALTEN Opened This Subject

The Smart Green Supply Chain programme had, by the start of this placement, built a credible set of components: a routing module, a supplier risk scorer, a carbon quantification module, a transport interface, and the DISCO twin that ties them together. Each answers a question of the form *what is the state of this part of the chain?*

None answers the question that comes next. Once the state is known, a coordinator still has to decide where to spend a finite amount of attention this week, and that decision is made on a signal none of those modules produces: how urgent each node is. In practice that signal comes from people. A buyer says a supplier is critical; a plant says a line is at risk. The digital twin records the consequence of those judgments without ever evaluating the judgments themselves.

That is a gap with a cost attached. Behavioural research on urgency shows the signal is not neutral. Zhu, Yang and Hsee demonstrate across five controlled experiments that people systematically favour imminent low-value tasks over distant high-value ones, and that the bias persists even when they are told about it (Zhu et al., 2018). Field studies find the same pattern under workload: operators complete small easy tasks first, at a measurable cost to overall performance (Rusou et al., 2020; DiwasSingh et al., 2020). If a digital twin consumes a human urgency declaration as ground truth, it inherits that bias and then propagates it across the network.

3.2 The Cost of the Two Errors

The two directions of error are not symmetric, which is what makes the problem worth instrumenting rather than tolerating.

Over-declaration is expensive but visible. A node declared more urgent than it is attracts expedited transport, premium capacity and coordinator attention, all of which are finite and all of which were taken from somewhere else. In a shared or mutualised logistics setting the cost lands on another actor entirely (Montreuil, 2011; Li et al., 2020).

Under-declaration is cheaper in the moment and far more dangerous. A node whose real situation is worse than anyone has said generates no alert, appears in no priority list and consumes no attention. The failure surfaces only when it becomes a missed delivery, by which point the window in which it could have been prevented has closed. Clinical triage, where this asymmetry has been studied for decades, reaches the same conclusion: under-triage costs a life where over-triage costs a queue (Kim et al., 2020; Mackway-Jones et al., 2013).

3.3 The Problem, Stated

Problem statement.

A supply chain digital twin can compute the physical state of every node, and it can record the urgency a human declares for each of them. It does not compare the two. The declared signal is known to be biased, the computed signal is available, and the difference

between them is exactly the quantity that would tell a coordinator which node is being mismanaged rather than merely which node is struggling.

Can the gap between declared and computed urgency be measured reliably enough, inside a working digital twin, to be acted upon rather than merely observed?

The word *reliably* carries the weight. Producing a number that expresses the difference between two other numbers is trivial. Establishing that the number means something, that it predicts outcomes better than the raw indicators a coordinator already has, and that it is not simply measuring its own inputs, is the actual **engineering problem**. Section 7 shows that this distinction is not academic: the system produced the number without difficulty and then failed the reliability test, for a reason that took a dedicated experiment to identify.

4 The Mission

4.1 The Assignment as Written

The placement agreement defines the subject as *Stage Innovation: Ingénieur Supply Chain*, and the assignment as the update and development of a supply chain digital twin. The signed scope covers five workstreams: defining the twin's elements, meaning factory, storage, transport and cargo; updating its evaluation elements, meaning score, cost, quality, delay, CO₂ and planning; updating its evolution elements, meaning positive and negative hazards and change scenarios; integrating complexity through multiple supply chains and tier-2 and tier-3 subcontractors; and integrating Physical Internet elements. It further specifies that the work is an innovation project including a state of the art, formalisation of the research in a scientific report, multidisciplinary project management, and an evolutionary approach based on Agile and Test and Learn.

That scope is deliberately broad, which is normal for an R&D assignment and is itself a managed risk (Section 5.5). The first weeks were therefore spent converging it onto a defensible contribution, in agreement with the research programme director.

4.2 The Converged Objective

The evaluation workstream is where the twin was weakest, because it computed state without ever assessing the human judgment applied to that state. The mission converged on that workstream, and specifically on the urgency dimension of it.

Converged objective. Design, implement and empirically evaluate a decision-support layer that computes a real urgency from the twin's operational state, captures the urgency declared by the operator through a structured elicitation, propagates both across the supplier network, and scores the gap between them in a way that can be tested against recorded outcomes.

Two boundaries were set explicitly and held. The layer is built as a standalone application running alongside DISCO rather than inside it, because coupling to a Java and Angular platform under active development by another team would have made the schedule dependent on theirs. And the layer targets urgency alone, not the scheduling decision that follows from it, because a prioritisation signal that cannot be shown to be trustworthy has no business driving a schedule.

4.3 Objectives, Indicators and Deliverables

Table 1 states the objectives agreed with the programme director, each with the indicator used to judge it and the deliverable that carries it. The indicators were chosen so that each could be answered yes or no from an artefact rather than from an opinion.

Intermediate deliverables. Four were produced during the mission rather than at its end: a knowledge transfer report on planning and urgency, delivered in week 2 and used to align vocabulary with the programme; a state of the art with an accompanying presentation, delivered in week 3; the mathematical specification, reviewed before implementation began;

Table 1: Objectives, indicators and deliverables

	Objective	Indicator	Deliverable
O1	Define a real urgency computable from heterogeneous operational indicators, and an adequacy score against the declared value	A model whose every term is decomposable, and whose aggregation choice is justified against the alternatives	Mathematical specification
O2	Deliver a working, persistent, auditable application implementing it	Runs multi-tenant; every write audited; automated test suite passing	SupplyScore application
O3	Propagate urgency across a multi-tier network and rank nodes by criticality	Handles a network at the scale of a real programme within an interactive time budget	Propagation and criticality services
O4	Establish whether the resulting signal predicts recorded outcomes	A score against a stated outcome definition, compared with a trivial baseline	Pre-registered experimental protocol and its results
O5	Capitalise the research so the programme can continue it	A written scientific record and reproducible derivation of every number	<i>Bilan Scientifique et Technique</i> and derivation scripts

and the frozen experimental protocol, hashed and registered before any campaign data existed, so that the hypotheses could not be adjusted after seeing results.

Final deliverables. The SupplyScore application with its test suite and documentation; the experimental campaign with its two executed arms and their analysis; the *Bilan Scientifique et Technique*; and the derivation scripts that regenerate every reported figure from raw artefacts.

The fourth objective became the centre of the mission. O4 as agreed asked whether the resulting signal predicts recorded outcomes, and its indicator required the comparison to be made against a trivial baseline rather than in absolute terms. Holding to that clause is what turned a validation step into the main scientific contribution of the placement. It produced three tests aimed at the system rather than for it: a scoring against recorded outcomes that failed and named its own cause (Section 7.4); a second supply chain specified to be hostile to the first result (Section 7.6); and the baseline comparison itself, which asks whether the most expensive component of the system earns its cost (Section 7.6.4). Section 8.1 states what each returned.

5 Project Organisation

5.1 Stakeholders and Responsibilities

The mission was carried out inside a small research team, and the distribution of responsibility mattered more than its size suggests. Table 2 lists who did what.

Table 2: Stakeholders and their role in the mission

Person	Role	Responsibility in this mission
Sébastien Perthuisot	Research Programme Director, Quality & Supply Chain Expert	Set the research direction, arbitrated scope, chaired the weekly regular point and the Friday demonstration. Decision-maker on what the programme keeps.
Émilienne Lardy	Scientific supervision	Ran the Monday planning and the three weekly scrums; reviewed method and written output. First reader of anything claimed.
Selma Ferhat	Scientific supervision	Reviewed the scientific framing and the written record.
Fabien Jules	Chargé Innovation	Placement tutor of record; onboarding, administrative framing and access.
Liam Jones	Fellow intern, same programme	Network representation, graph modelling and geographic layer of the twin. Neighbouring problem, shared demonstration slot.
David Gomez	ESTIA academic supervisor	Academic framing of the mission.
Author	Intern, Supply Chain Engineer	Model, implementation, experimental design, analysis and written deliverables.

The presence of a second intern on an adjacent problem shaped the work more than expected. Liam Jones was building the structural and geographic representation of a supply chain, meaning how a network is described, generated and visualised, while this mission built the layer that scores nodes once they exist. The two are complementary halves of the same twin, and the Friday demonstration forced each to state weekly what the other could rely on. It also produced a boundary decision early: the scoring layer would assume a network is given rather than attempt to generate one, because that was being solved two seats away.

5.2 Resources

The resources provided were modest and are worth stating plainly, because two of them constrained technical choices.

- *Human.* No budget-holding team. Supervision as listed above, and one peer on an adjacent subject. All implementation was carried out by the author.
- *Material.* One laptop provided by ALTEN, a Ryzen 7 PRO 8840HS with 16 cores and no discrete GPU. Everything computed in this mission, including every model training run, ran on that CPU.
- *Software.* Azure DevOps for source control and work tracking, and the Microsoft suite for communication. No paid data service, no cloud compute budget, no commercial

solver or licence was requested or used, which is why the delivered system depends only on open-source Python libraries and a file-based database.

- *Financial.* None directly managed. The absence of a budget is itself a design constraint: it removed managed cloud services and paid datasets from the solution space before any technical comparison took place.

5.3 Working Method: Agile and Test and Learn

The placement agreement prescribes an evolutionary approach based on Agile and Test and Learn. That was applied literally rather than nominally, and the weekly rhythm was the mechanism.

The cadence. The week had a fixed shape. A planning session on Monday set the objective for the week. A scrum on Tuesday, Wednesday and Thursday morning kept it visible and surfaced blockages within a day rather than a week. A regular point with the programme director on Wednesday took the scientific decisions. A demonstration on Friday closed the loop by requiring something to be shown, not described. Four contact points a week on a six-month mission is a heavy cadence for R&D, and it was the single most useful organisational feature of the placement: it made it impossible to spend three weeks on an idea that would not survive contact with a reviewer.

Test and Learn, with what it actually rejected. The method is only meaningful if things were genuinely abandoned. Four were.

1. *The first urgency formulation was dropped.* The initial model expressed urgency as a single time-driven function with a finite-time singularity, calibrated by penalty terms. It was specified, partially implemented and abandoned in favour of a multi-block aggregation, because a single deadline-proximity curve discards information the twin already holds and cannot represent a node that is on time but out of capacity.
2. *The May module architecture was replaced.* A first pipeline of separate scores was built through May and superseded in June by the aggregation and propagation design that shipped. The intermediate work was not wasted: it established what the interfaces between components needed to be, which is why the replacement was built in weeks rather than months.
3. *A candidate dataset was rejected on inspection.* A third-party logistics dataset considered for the freight node was checked before use and discarded: it carried pre-computed target columns, coordinates scattered outside the region it claimed to cover, and uniformly flat distributions, all signatures of synthetic data. The node was driven by calibrated events instead. The rejected file is retained for audit and never loaded by the analysis pipeline.
4. *A whole experimental arm was reduced.* The campaign was designed for eight human consultants over nine working days. When that could not be secured, the design was changed rather than the claim: the campaign was run with isolated agents, and every conclusion about human behaviour was correspondingly weakened in the reporting.

Figure 2 shows the loop as it was actually run. Each turn of it produced either a working increment or a documented rejection, and the four rejections above are the evidence that the second outcome was genuinely available rather than theoretical.

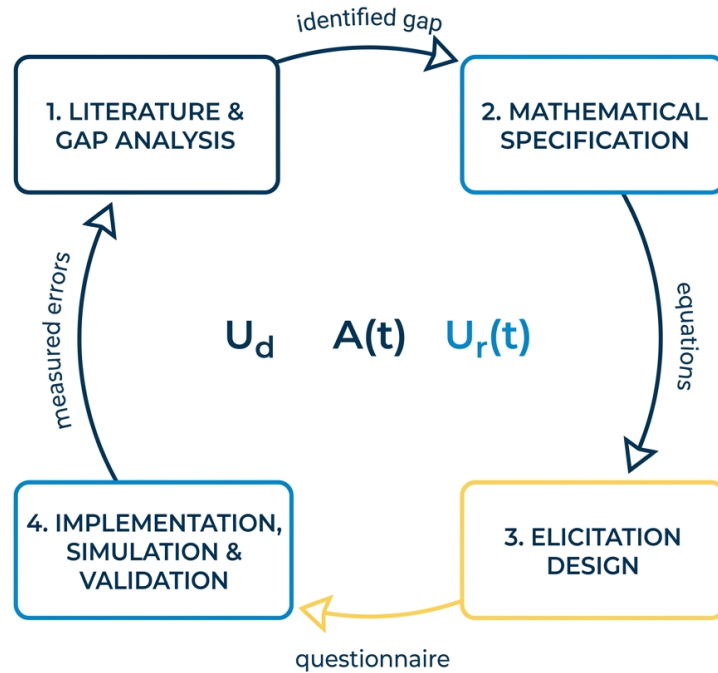


Figure 2: The test-and-learn iteration loop

Quality gates. Agile without a definition of done produces velocity and nothing else. Three gates were applied: no feature merged without automated tests, giving 1779 test functions at the end of the mission; every business write passing through a single audited mutation service, so that no result could be produced by an untraceable edit; and every reported figure regenerated from raw artefacts by a script rather than transcribed by hand.

5.4 Phases and Milestones

The mission ran in six phases. Figure 3 gives the schedule and Table 3 the milestones that closed each phase.

Figure 3 marks the point this report is written from. Two phases are complete beyond the campaign that was originally the mission's end point, and the last two months are allocated to widening the evidence base rather than to new construction.

The planning was revised once, in June. The original schedule allocated the second half of the mission to extending the model's scope; it was reallocated to validating what already existed, on the argument that an unvalidated extension of an unvalidated model compounds the problem rather than advancing it. That reallocation is the reason this report can state a measured result at all.

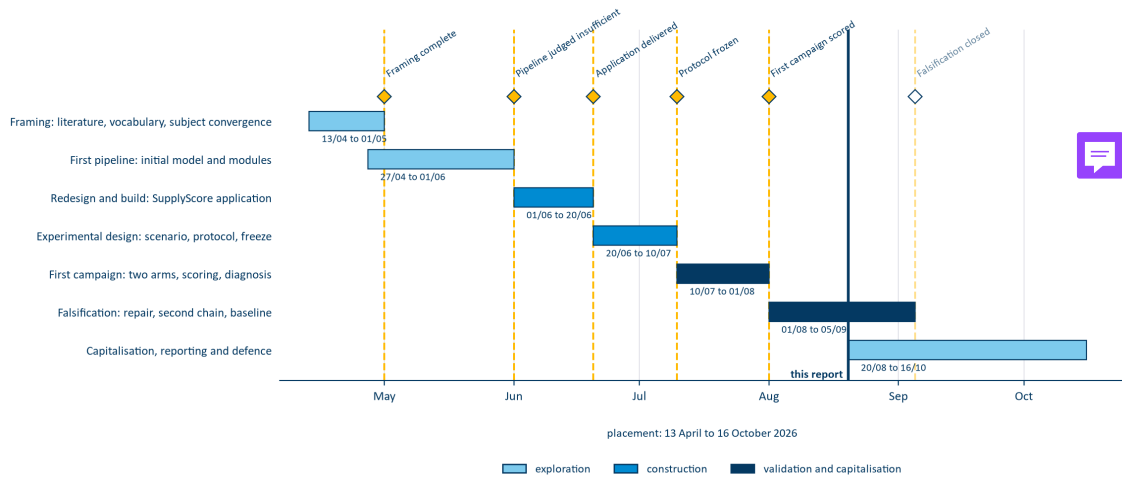


Figure 3: Phases and milestones of the mission

Table 3: Milestones

Date	Milestone	Closed by
End April	Framing complete	Knowledge transfer report and state of the art presented; subject converged with the programme director
End May	First pipeline	End-to-end computation running, and found insufficient, which triggered the redesign
Mid June	Application delivered	Multi-tenant persistence, audited writes, weekly cycle, event engine, explainability, exports and backup
Early July	Protocol frozen	Scenario, data pack hashed, role cards and hypothesis register registered before any data existed
End July	First campaign scored	Two arms run and analysed against the pre-registered register; the forecast layer's defect identified and located
Mid August	Falsification	Repair re-measured on frozen forecasts; second chain built and run in both arms; baseline comparison scored
Early Sept	Falsification closed	<i>Planned</i> . Remaining chains run and the sample base widened

5.5 Risks and Mitigations

Four risks were identified and carried at the start of the mission. Table 4 states each one, whether it materialised, and what was done about it. Three of the four did materialise, which is a high rate and reflects how much of this mission depended on things outside the author's control.

One consequence of the first risk is worth recording for whoever continues the work. The state of the art rests on sources retrieved under a student affiliation, so the reading list was capitalised alongside the code: every retained source is registered with its argument, its method and its limits, and the register is a deliverable rather than a working note. The next holder of the subject inherits the appraisal even where they cannot re-download the paper.

Table 4: Risk register and outcomes

Risk	Outcome	Mitigation and effect
Literature access routed through university libraries	Materialised	The state of the art was built on ESTIA and Cranfield library credentials, and retrieving a paper was never a same-hour operation. Mitigated by front-loading retrieval into the first three weeks against a planned list of the sources later phases would need, so that no design decision ever waited on a download.
Broad mission scope	Materialised	Five workstreams in the signed assignment against one person and six months. Mitigated by converging the subject in the first weeks with the programme director and recording the boundaries explicitly (Section 4.2) rather than leaving them implicit.
Human campaign might not be securable	Materialised	Eight colleagues for nine working days was never guaranteed. Mitigated by designing the protocol so that the declarant could be substituted without changing the record format, which is what allowed the campaign to run at all when the humans were unavailable. Cost: the external validity of every behavioural claim.
CPU-only compute	Contained	No GPU available. Bounded model size and training resolution, and is the reason one comparison was run at reduced scale. Mitigated by choosing methods whose cost is dominated by data rather than by parameters.

6 State of the Art

The question of Section 3.3 sits at the meeting point of several fields that rarely cite one another. This section reviews each for what it settles and, more usefully, for what it leaves open. A structured appraisal of every source, recording its argument, its method and its limits, was maintained throughout the mission and informs the readings below.

6.1 Deadline-Driven Value and Risk Propagation

Real-time systems research models the value of a task as a function of when it completes. Chen and Mühlethaler show that utility typically holds constant until a deadline and then decreases (K. Chen et al., 1996), and a wider survey confirms structured degradation past critical thresholds (J. Chen et al., 2025). This establishes that urgency is legitimately time-varying.

Separately, supply chain research formalises the ripple effect, whereby a local disruption cascades through downstream tiers and amplifies as it travels (Ivanov; Dolgui, 2020; Ivanov; Dolgui, 2019; Li et al., 2020). This establishes that urgency is legitimately network-dependent.

Appraisal. The first stream stops at a single isolated task and offers nothing on combining heterogeneous indicators. The second models propagation but treats the node-level score as given. Neither addresses a declared signal at all. Between them they justify treating urgency as time-varying and propagated, and leave the aggregation question entirely open.

6.2 Indicators of Delivery Rupture

If urgency is to be aggregated, the indicators must come from somewhere. Heckmann, Comes and Nickel establish rupture as the terminal event a supply risk measure should anticipate, and criticise the field for measuring exposure without connecting it to that event (Heckmann et al., 2015). Indicator-based supplier monitoring proceeds by identifying observable precursors (Mokhtar et al., 2019).

Five families recur. Slack is the only indicator whose exhaustion is irreversible (Cohen et al., 2019; Sun et al., 2020). Capacity saturation adds waiting time a nominal lead time does not show (Tomlin, 2006; Hosseini; Tajik, et al., 2019). Resource performance is monitored through composite proxies because supplier-side data is rarely available (Mokhtar et al., 2019; Hlioui et al., 2017). Adverse-event exposure is treated probabilistically (Hosseini; Ivanov, et al., 2020; Liu et al., 2021). Cost signals are read as symptoms of strain rather than causes (Shen et al., 2019), and emissions constraints restrict the admissible responses rather than causing rupture (Hoen et al., 2014; Smart Freight Centre, 2023).

Appraisal. This literature names precursors and monitors them individually. It supplies no criterion for deciding which belong in one composite, and says nothing about whether equal numerical values on structurally different indicators carry equal risk meaning. Both questions are inherited by any multi-block model, and Section 7.1 treats the second explicitly.

6.3 Cognitive Bias in Declared Urgency

Zhu, Yang and Hsee report the mere-urgency effect across five controlled experiments: participants consistently preferred imminent low-value tasks to distant high-value ones, and the bias survived being told about it (Zhu et al., 2018). Steel gives a formal discounting model in which immediacy becomes more salient as a deadline approaches (Steel et al., 2006). Field studies confirm the pattern under workload (Rusou et al., 2020; DiwasSingh et al., 2020).

Appraisal. This stream diagnoses the problem and quantifies it in controlled settings, but the magnitude in industrial supply chains is uncalibrated, and it prescribes no operational correction. It justifies treating the declared signal as biased; it does not say what to do about it.

6.4 Multi-Criteria Elicitation and Aggregation

Structured elicitation of a human judgment is a solved problem in outline. The Analytic Hierarchy Process derives criteria weights from pairwise comparisons and validates them through a consistency ratio (Saaty, 1980), though it is vulnerable to rank reversal and to cognitive noise (Dyer, 1990; Triantaphyllou et al., 2024). The Best-Worst Method reduces the number of comparisons required (Rezaei, 2015), and its fuzzy extension absorbs judgment uncertainty (S. Guo et al., 2017; Liang et al., 2020). For ranking alternatives without letting a good score on one criterion mask a catastrophic one on another, outranking methods such as PROMETHEE II establish pairwise preference flows (Brans et al., 1986; Behzadian et al., 2010).

The composite indicator literature supplies the necessary warning: nominal weights often diverge from effective importance because of correlation and normalisation effects (Greco et al., 2018; Becker et al., 2017; Borgonovo et al., 2021).

Appraisal. These are static tools. They elicit and aggregate, and none is routinely compared against a computed operational counterpart. The composite-index critique is generally raised as a caveat at the end of a paper rather than turned into something a running system checks on its own data.

6.5 Digital Twins and Decision Support

The digital twin concept was formalised by Grieves (Grieves, 2014) and extended to manufacturing (Tao et al., 2018; Fuller et al., 2020). In supply chains it integrates simulation, optimisation and machine learning (Ivanov; Dolgui, et al., 2021), with demonstrated capability in disruption monitoring and replanning (Ding et al., 2023; Leung et al., 2022; Hrušovský et al., 2021). Recent human-in-the-loop cognitive twins capture human feedback to synchronise mental models with simulated operations (Niloofar et al., 2023).

Appraisal. This is the family the delivered system belongs to, and the gap is here. No retrieved system combines a declared urgency input, a computed operational urgency and their propagated mismatch. A second gap matters for this mission specifically: these systems are evaluated on how accurately they estimate physical state, and their inputs are assumed to arrive. Where an input depends on the same operator whose judgment the system exists to support, that circularity is not treated as a measurement problem.

6.6 Industrial Solutions and Patents

Academic absence would be weak evidence on its own, since a working technique can be commercialised without ever being published. The patent record was therefore searched for the same combination.

The earliest sustained line is IBM's, which quantifies revenue at risk across quantity, time, location and quality dimensions of a disruption. Its inputs are the bill of materials, supplier locations, component costs, product pricing, demand volatility, strike probability and lead times, and the risk distribution is produced by simulation (Corporation, 2007). Every input is a measured or forecast quantity; no operator states an opinion anywhere in the claims.

The second line replaces measured inputs with language. Wipro's method takes a natural language description of a problem, extracts entities, maps them onto pre-declared supply chain component clusters, and derives association rules linking an identified risk to probable subsequent ones (Limited, 2019). A human is present, but as the narrator of an incident rather than as an assessor whose judgment is itself evaluated; the text is a source of facts, and nothing in the method compares it against a computed counterpart.

The third and most recent line is the closest to the environment this mission works in: a digital twin control tower claiming diagnosis of delay sources and prediction of maintenance needs across replicated value chain entities (LLC, 2022). Its orientation is automation, with robotic process automation and adaptive intelligence acting on the twin; the operator is a recipient of decisions rather than a measured party to them.

Appraisal. The three lines trace a clear trajectory over fifteen years, from measured inputs to unstructured text to autonomous action, and it runs in one direction: away from the human declarant. Prior art protects how to compute risk and how to act on it. None of the retrieved claims covers eliciting a declared urgency, computing its operational counterpart and scoring the difference, which is the combination this mission builds. Commercial offerings in the same space, whose public documentation describes risk scoring and event alerting rather than declaration quality, are consistent with that reading.

6.7 Verifying a Probabilistic Signal

Because this mission scores a probabilistic signal against outcomes, it inherits forecast verification. Brier introduced the mean squared error of probabilistic forecasts (Brier, 1950), and Murphy decomposed it into reliability, resolution and uncertainty (Murphy, 1973), establishing the distinction the results of this report depend on: a forecast can order outcomes correctly while announcing the wrong probabilities, and the two failures need different repairs. Machine learning rediscovered the same point, finding strong classifiers that produce distorted probabilities correctable by a cheap post-hoc layer (Niculescu-Mizil et al., 2005; C. Guo et al., 2017).

A second convention is inherited with it. A verification score is read against a reference forecast, not in absolute terms, and the reference is chosen to be the cheapest thing a practitioner could have done instead: climatology, persistence, or a one-line heuristic (Murphy, 1973). Section 7.6.4 applies that convention to the layer built here, and it is the most demanding test in this report.

Appraisal. This literature assumes the outcome being predicted is unambiguously defined. In

an operational supply chain it is a modelling choice, and Section 7.4.5 measures that choice moving the reported result by a factor of six.

6.8 What the Review Establishes

Each stream closes on a limit, and the limits describe one absence rather than several. Figure 4 places them side by side.

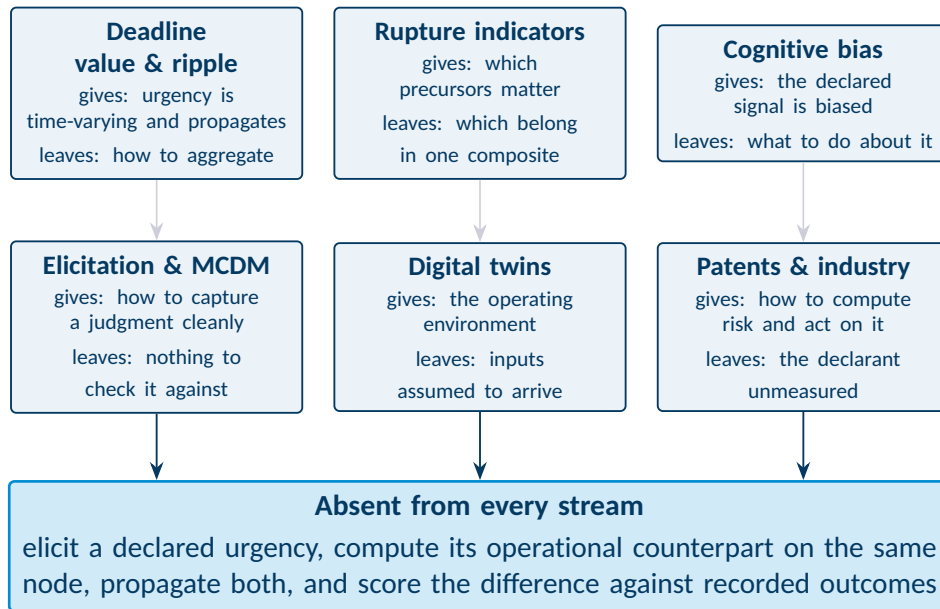


Figure 4: Six streams, what each settles and what each leaves open. The lower box is the combination none of them provides, and the one this mission builds.

The reviewed work covers multi-block risk aggregation, network propagation, digital twin operation, multi-criteria elicitation, industrial risk scoring and probabilistic verification. No retrieved work, published or patented, combines them with a structured declared-urgency input and an empirical test of the resulting adequacy signal against recorded outcomes. That missing combination is what this mission set out to build and, more importantly, to try to break.

7 Carrying Out the Mission

This section follows the phases of Section 5.4. The model comes first, then the system built from it, then the three tests run against that system: a scoring against recorded outcomes and the repair it triggered, a second supply chain built to break the result, and a comparison against the cheapest predictor that could have replaced the most expensive component. The problems met along the way close the section.

7.1 The Model

7.1.1 Real urgency

Real urgency is not read from a single deadline-proximity signal. It is aggregated at each node from six indicator blocks selected from the literature of Section 6: time, capacity, performance, risk, cost and carbon. Each produces a partial urgency $u_m \in [0, 1]$, and the aggregation is a weighted probabilistic OR:

$$U_{r,\text{local}} = 1 - \prod_m (1 - u_m)^{\omega_m}. \quad (1)$$

The choice of that form over a weighted average is the central modelling decision. A weighted average is fully compensatory: a node with a completely failed capacity and healthy cost, carbon and performance readings would score as moderately urgent. Equation 1 is not: any block approaching its maximum drives the aggregate toward the maximum regardless of the others. For a signal whose purpose is to catch the node that is about to fail, non-compensatory behaviour is the required property.

Table 5: The six indicator blocks and what each measures

Block	Role	What it measures
Time	Direct cause	Probability that the realistic lead time exceeds the margin to the next deadline. The only resource whose exhaustion cannot be reversed
Capacity	Direct cause	Saturation of storage, weight and throughput, which adds waiting time a nominal lead time does not show
Performance	Direct cause	Drift in the productive health of the resource making the part, through a composite availability, performance and quality proxy
Risk	Direct cause	Expected unavailability: failure probability times recovery time times severity, revised weekly from declared events
Cost	Arbitration	Financial strain. Paying premium freight to protect a deadline is a symptom, not a cause, and it bounds which responses stay affordable
Carbon	Arbitration	Emissions above target. Restricts the admissible set of responses, typically in the sea-against-air trade-off

Table 5 gives the six and separates them by role. Four describe a mechanism that leads to rupture; two constrain which responses remain available once it threatens. Each candidate indicator was screened against four questions before being retained: does it have a plausible causal link to rupture, can it be measured or credibly estimated, can an operator act on it, and

does it add information the retained blocks do not already carry. Figure 5 draws the resulting structure.

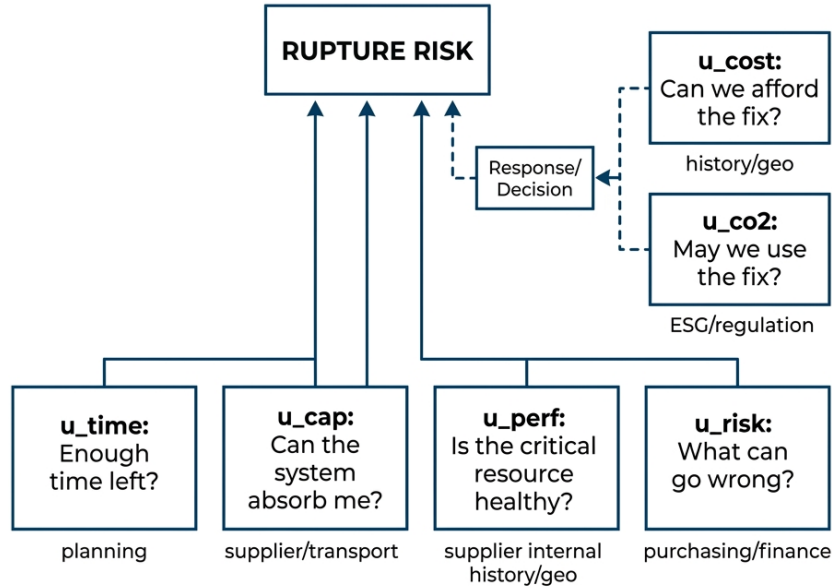


Figure 5: Causal structure behind the six-block selection

Three properties of that equation were specified explicitly, because a bare formula does not state what its bounds mean. At $u_m = 0$ the block is nominal and its factor is 1, leaving the product unchanged. At $u_m = 1$ the factor is 0 and the aggregate saturates. A block whose data is unavailable is given weight $\omega_m = 0$, which is deliberately distinct from $u_m = 0$: the first says nothing is known, the second asserts the block is healthy.

The assumption that the blocks are exchangeable. The product is symmetric in its factors, so it treats the six blocks as interchangeable up to their weights. That does not hold cleanly. A value of $u_{\text{time}} = 0.3$ is a probability of missing a deadline; $u_{\text{cost}} = 0.3$ is an ordinal position on a normalising scale wearing a cardinal number. They are not equally informative about rupture. The consequence was carried rather than hidden: the aggregate is used for ranking and for comparison against the declared value, never as a calibrated probability in its own right.

7.1.2 Declared urgency and the adequacy score

Declared urgency is not collected as a free-text estimate. At each node the operator compares four criteria pairwise through AHP, which yields a consistency-checked weight per criterion,

then rates each on a severity scale. The four criteria were chosen so that each has a computed counterpart: operational impact and downstream dependencies mirror the severity the propagation amplifies, the time window mirrors the time block, and recoverability mirrors the recovery term of the risk block. Without that correspondence the later comparison would be between unrelated quantities.

Figure 6: The pairwise comparison questionnaire and its consistency gate

Figure 6 shows the questionnaire as the operator meets it. A weight vector whose consistency ratio exceeds the accepted threshold is refused, and the operator is sent back to the most contradictory of their own comparisons before the assessment can enter the network. Without that gate an incoherent judgment would propagate as though it were a considered one.

Both signals then propagate across the supplier network, and in opposite directions: declared urgency descends from the customer toward suppliers, real risk ascends from suppliers toward the customer. The asymmetry is deliberate. Demand pressure is something a customer transmits to its suppliers; physical risk is something suppliers transmit to their customer. A model that propagated both the same way would be describing neither.

Figure 7 shows the two signals travelling in opposite directions across the same graph.

Their mismatch, once both are propagated, splits into two named pathologies: false urgency where the declaration exceeds the state, hidden risk where the state exceeds the declaration. It is scored on $[0, 100]$ through an asymmetric penalty that prices hidden risk at roughly twice false urgency, following the clinical principle of Section 3.1.

Figure 8 shows the two signals over the life of a task. The step line is the declaration, which moves only when a review is submitted; the smooth curve is the computed state, which moves continuously. Where the step line sits above the curve the enclosed area is false urgency; where the curve sits above the step line it is hidden risk.

7.2 The System

The model was delivered as SupplyScore, a standalone Python application of 73 modules across 10 packages, covered by 1831 automated test functions in 106 test files, driving a multi-page web interface over a per-node database layer. The suite collects 2 044 cases once parameterisation is expanded, of which 2 028 pass, 15 are skipped for want of an optional

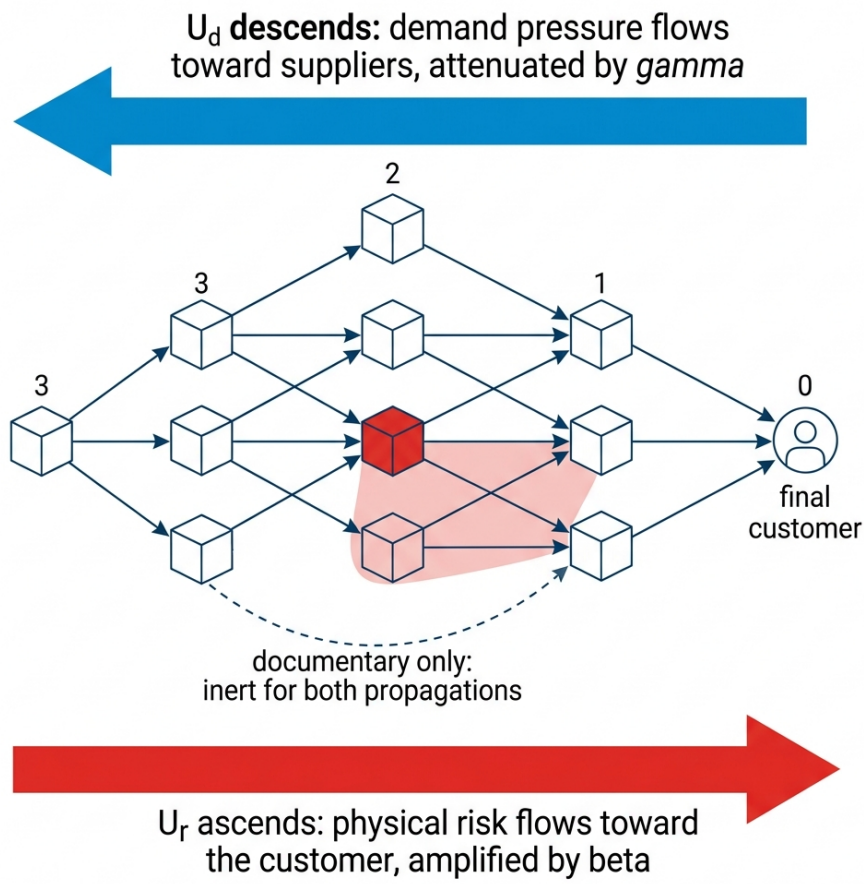


Figure 7: Bidirectional propagation across the supply network

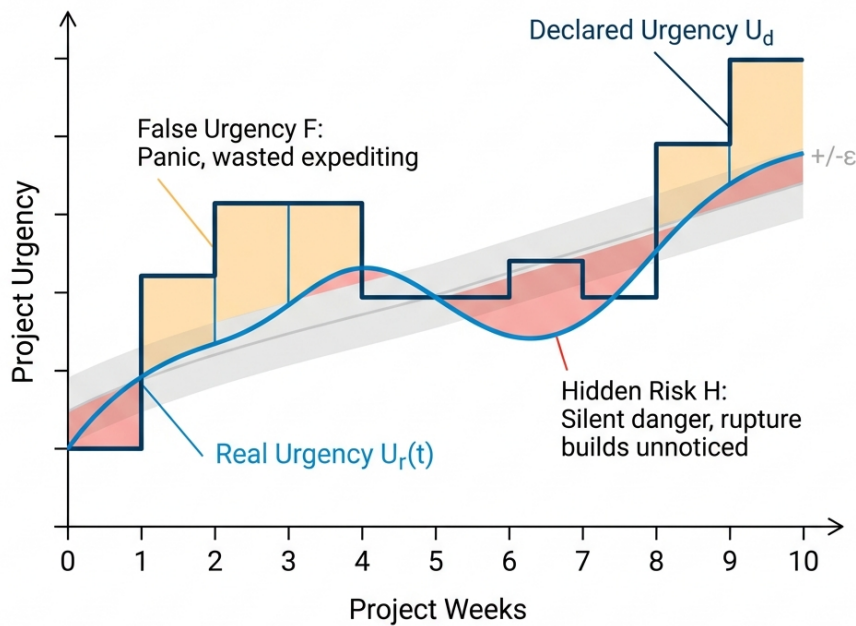


Figure 8: Declared and real urgency, and the two mismatch pathologies

dependency, and one fails on two missing screenshot files referenced by a presentation script outside the library.

Four architectural decisions carry the engineering weight. Every operation passes through a single orchestrating facade, so there is no second write path. Data is split into one registry database plus one database per node, which enforces that one supplier's history is not readable through another's. Every business write crosses a single mutation service that validates ranges, diffs against current state and records the change with its author, making the audit trail a property of the architecture rather than a convention. And each project carries its own clock, which is what allows a six-month scenario to be replayed in compressed time without touching the model.

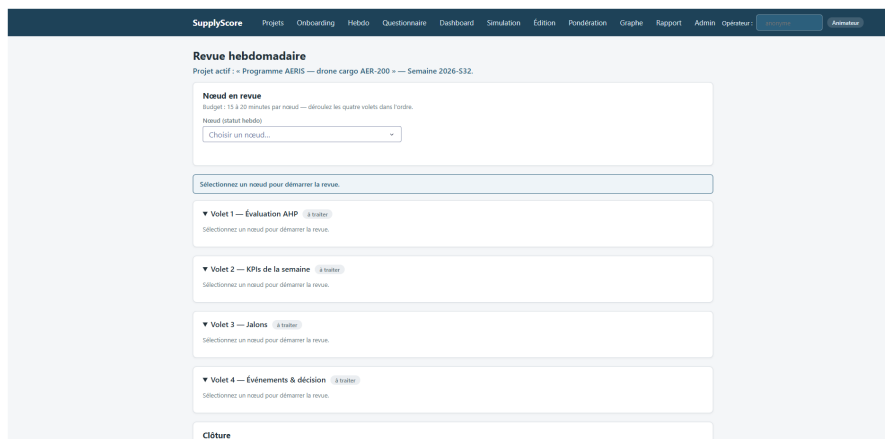


Figure 9: The weekly review page

Figure 9 shows the operator's main entry point. Its four sections, the questionnaire, the week's indicator values, milestone progress and any events, commit as a single signed transaction so that no partial review can enter a node's history.

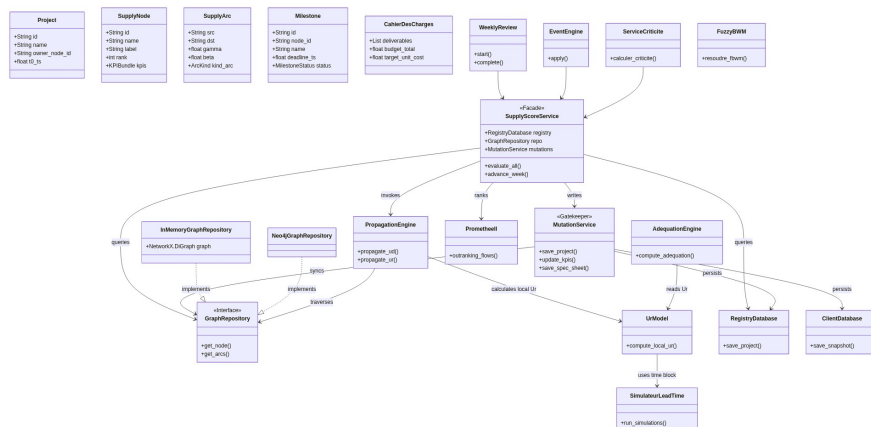


Figure 10: System architecture: single facade, audited writes, per-node persistence

Figure 10 shows how those pieces relate. The single entry point is what makes the audit guarantee enforceable rather than aspirational: there is no path into the data that bypasses it, including from the administration interface.

Two capabilities proved more important than anticipated. The explainability page reconstructs any score term by term, decomposing the declared value into one contribution per criterion

and the computed value into one per block, so that a coordinator asking why a node scored as it did receives an exact answer rather than a plausible one (Figure 11). That is what made the diagnosis in Section 7.4 possible from the application rather than by hand. The export produces frozen data for analysis outside the tool, which the whole experimental analysis relies on.

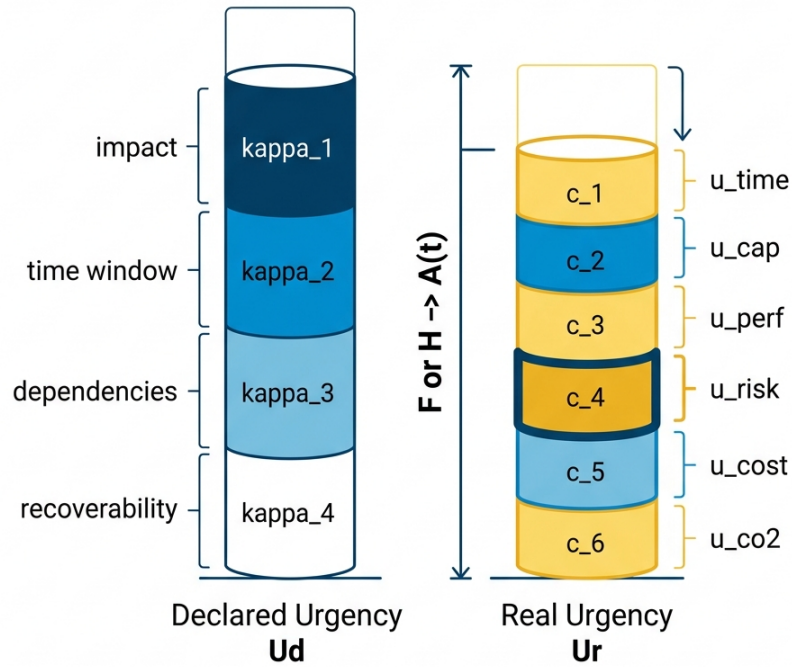


Figure 11: Per-node explainability: exact decomposition of both signals

7.3 The First Campaign

Building a system that computes a number establishes nothing about whether the number means anything. The mission therefore included an experiment designed before the system was finished.

The campaign replays the 2020 to 2022 semiconductor shortage over nineteen turns across an eight-node satellite supply chain, from a raw wafer supplier through a foundry, a distributor, a freight operator and an electronics manufacturer to the satellite prime. Real urgency is fed from public time series of the actual crisis; declared urgency comes from the participants through the weekly questionnaire. The two signals therefore have independent sources, which is the precondition for the gap between them to mean anything.

Figure 12 gives the network, with the coefficient attenuating demand as it descends and the one amplifying risk as it ascends, and each node's tier. The foundry at the deepest tier is the expected bottleneck. The dashed arc into the secondary foundry is deliberately inert: it represents the announced alternative supplier that in the real crisis had no spare capacity, and modelling it as present but carrying nothing is the point.

Four artefacts were frozen before any data existed: the scenario with its twelve calibrated events, the data pack with a hash recorded per file, the role cards, and a register of six

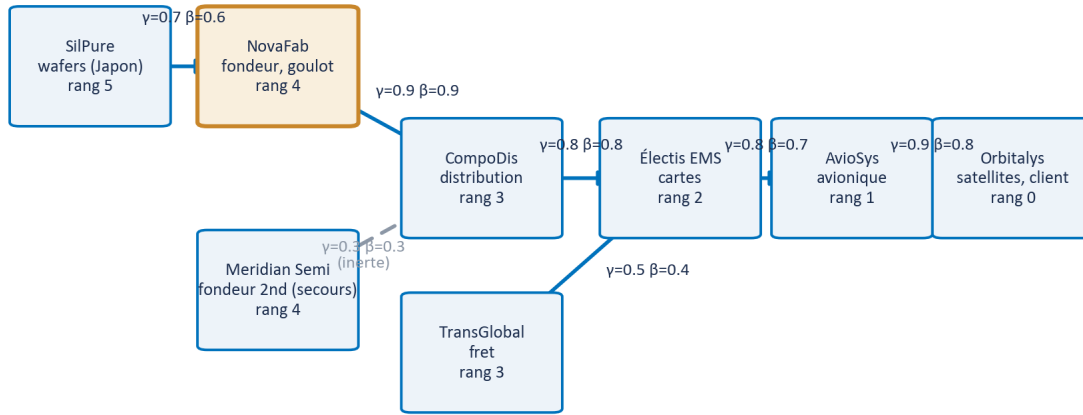


Figure 12: The campaign network and its per-arc coefficients

hypotheses each carrying a numeric threshold. Freezing the thresholds in advance is what prevents a disappointing result from being rescued by moving the bar afterwards.

The design compares two arms on the identical frozen scenario. In the control arm, eight isolated declarants each see only their own role card and turn briefing and declare blind. In the treatment arm, the same declarants are additionally shown the system's own predictive analysis of their node before declaring, and record whether it influenced them. Because the scenario is frozen, the forecasts are identical between arms, so any behavioural difference belongs to the act of displaying them.

7.4 Results on the First Chain

The results divide into what the system was built to do, which it does, and the predictive layer built on top of it, which was diagnosed, repaired and measured as improving without yet being good. They are reported in that order, and both are then re-examined on a second chain in Section 7.6.

7.4.1 The adequacy signal identifies the right nodes, and does it early

The first result is the one the system was built for, and it is positive.

A coordinator does not ask whether node X will suffer an adverse event between weeks $t + 1$ and $t + 4$. They ask which of their suppliers deserves attention this week. That is a question about nodes, not about node-weeks, and the adequacy signal answers it. At each turn the eight nodes were ranked by hidden risk, the part of the gap where computed urgency exceeds declared urgency, and the set flagged above 0.10 was compared with the set of nodes that would go on to miss a milestone at its originally committed deadline. Four of the eight nodes did.

Table 6 gives it. For six consecutive turns, from the opening turn to the one before the foundry freeze, the signal flags exactly three nodes, all three of which go on to miss a commitment, with no false alarm at any turn. The fourth is never flagged early, so recall stops at three of four. And on every one of those six turns the worst-scoring node of the eight is the satellite prime, which misses its delivery deadline at turn 8 and delivers at turn 14: the flag precedes

Table 6: Nodes flagged by hidden risk against nodes that later missed a commitment

Turns	Precision	Recall	False alarms	Reading
T0 to T5, before the crisis is visible	1.00	0.75	0	Three nodes flagged, all three later missed
T6 to T18, crisis public	0.50 or below	0.25	1 to 3	The declared signal catches up and the gap closes

the miss by eight turns, and it is raised while that node's own manager is declaring an urgency of 0.001.

That last detail is the whole point of the construction rather than a happy accident. Nothing in the flag comes from the declaration being pessimistic; it comes from the declaration being optimistic while the computed state is not. A system that only aggregated indicators would have ranked that node by its indicators. A system that only collected declarations would have believed it. The gap between the two is what carries the information, which is what the mission set out to test.

The signal then degrades from turn 6 onward, and the reason is structural rather than a defect. Once the crisis is public every declarant raises their urgency, declared catches up with computed, and a signal defined as the difference between the two necessarily shrinks. Precision falls to 0.50 and recall to 0.25. This is the correct behaviour of a perception-gap measure and it bounds where such a measure is useful: it is an early-warning instrument, not a crisis-management one. Stated the other way round, the signal is at its sharpest exactly when a warning is still worth something.

The repair of Section 7.5 sharpens the same flag without changing which nodes it selects. On the campaign as run, the satellite prime scores 26.6 out of 100 at turn 3 against 46.0 for the next worst node; after repair the same turn gives 2.2 against 44.6. The ordering is identical and the separation is an order of magnitude wider, which is what makes the flag readable on a dashboard rather than merely present in the data.

One limit bounds all of this and is the reason Section 8.3 asks for more data before anything is generalised: eight nodes, four of which are positive, on one scenario. A precision of 1.00 over six turns on three flagged nodes is an encouraging observation and not an established rate.

7.4.2 The forecast layer, and the definition it is scored against

The adequacy signal ranks nodes. Turning it into a probability that a named outcome occurs within a named window is a further step, and that step is where the campaign found its defect. Scored against the outcome definition the system itself applies, meaning a missed milestone, an abandoned node or a critical event within four weeks, the forecast layer failed on both axes that matter.

Discrimination is indistinguishable from chance: the area under the ROC curve ranges from 0.37 to 0.56 across horizons and every confidence interval contains 0.5. That statement carries a caveat, since the positive class holds between two and six observations, which is too few to conclude in either direction.

Calibration fails much more clearly, and does not depend on that small sample. The Brier skill against a forecaster that simply announces the base rate is between -3.4 and -6.9 , meaning the model is several times worse than the trivial baseline. Table 7 localises the failure.

Table 7: Reliability of the four-week forecast

Announced band	<i>n</i>	Mean announced	Observed	Reading
[0.0, 0.1)	74	0.070	0.081	Well calibrated
[0.1, 0.2)	24	0.128	0.000	None materialised
[0.2, 0.5)	1	0.354	0.000	None materialised
[0.5, 0.9)	7	0.859	0.000	None materialised
[0.9, 1.0]	14	0.992	0.000	None materialised

The pattern is unambiguous, and Figure 13 shows it, with each marker’s area proportional to the number of forecasts in its band and the diagonal marking perfect calibration. Where the model announces a low probability it is right: 74 forecasts averaging 0.070 were followed by an adverse outcome 8.1% of the time. Where it announces a high probability it is wrong every time: twenty-one forecasts announcing a mean of 0.948 were followed by the predicted outcome zero times.

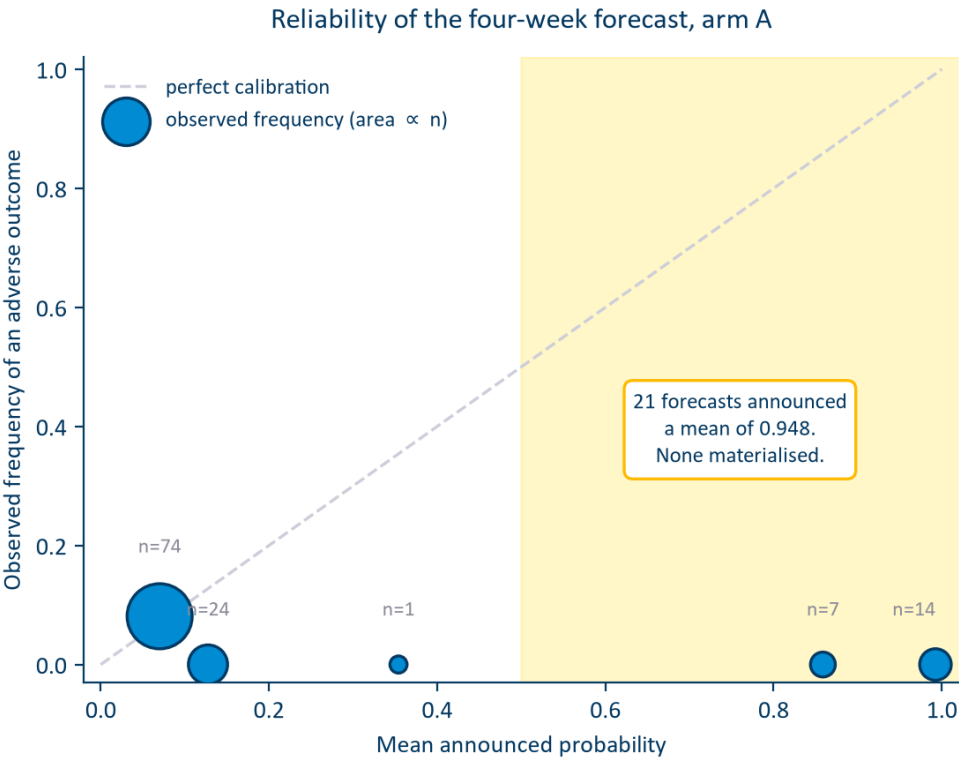


Figure 13: Reliability of the four-week forecast

Separating these two failures matters for what happens next, and it is the distinction Murphy established for exactly this purpose (Murphy, 1973). A forecast that orders outcomes correctly but announces the wrong numbers is repaired by a thin post-hoc layer that leaves the model untouched. A forecast that does not order outcomes at all needs the model changed. On

this campaign the evidence for the second is weak, resting on very few positive observations, while the evidence for the first is strong and does not depend on that sample at all.

7.4.3 Why: one block explains all of it

The explainability layer localised the cause. Across all twenty-one confident forecasts, the four-week probability and the milestone-miss probability never differ by more than 0.020: the confident tail is the milestone channel and nothing else. Set against the registry, ten of the eighteen campaign milestones completed, eight remained active and only one was still open past a deadline that had fallen inside the campaign.

That registry reading needs one qualification, because it is more forgiving than it looks. The registry holds each milestone's *current* deadline, and a replanning rewrites it, so a milestone re-dated and then delivered counts as completed. Scored instead against the deadline each node originally committed to, six of the eleven milestones whose original deadline fell inside the campaign were missed, five of them by a delivery that arrived one to six turns late. Under that stricter reading five of the twenty-one confident forecasts were followed by a missed original commitment within four weeks, rather than none. The confident tail is still wrong far more often than it is right, and the diagnosis below is unaffected, but the difference between zero of twenty-one and five of twenty-one comes entirely from which deadline the truth is read against. Section 7.4.5 returns to that.

Reading the time block directly at the turn the scenario's critical event lands confirms the diagnosis from the other side: the block reads numerically zero at six of the eight nodes in the middle of a crisis. The cause is structural. The block computes milestone risk from a lead-time distribution against a deadline, while the event engine writes to capacity, cost and failure probability and never to any quantity on the time axis. The strongest form of the evidence is a count: across the whole campaign, eight nodes and forty-one indicator snapshots each, the accumulated-delay field the block would need was written a non-zero value exactly zero times. No path connects a shock to the quantity the block reads, so the block cannot respond to a disruption. It is blind by construction, and because it drives the entire confident tail, so is the forecast.

This is a defect in the model rather than a property of the scenario. Any deployment would inherit it, and the repair is a change to the event mapping rather than to the mathematics. It is the single most actionable finding of the mission.

7.4.4 What happens when the forecast is shown to the operator

The treatment arm produced a second result, and one that was not sought.

For the first four turns the forecast is hidden and all eight declarants record no influence. At the turn it first becomes visible, all eight record one, and the direction is entirely one-sided: five revised their declared urgency downward and three had their existing assessment confirmed. Not one revised upward. The first upward revision anywhere appears the following turn, and the scenario's single critical event lands two turns after the moment every declarant was either reassured or confirmed.

Figure 14 shows the distribution turn by turn. Over the full campaign the balance is close to even, 15 upward against 16 downward, so the one-sided effect belongs to first contact rather

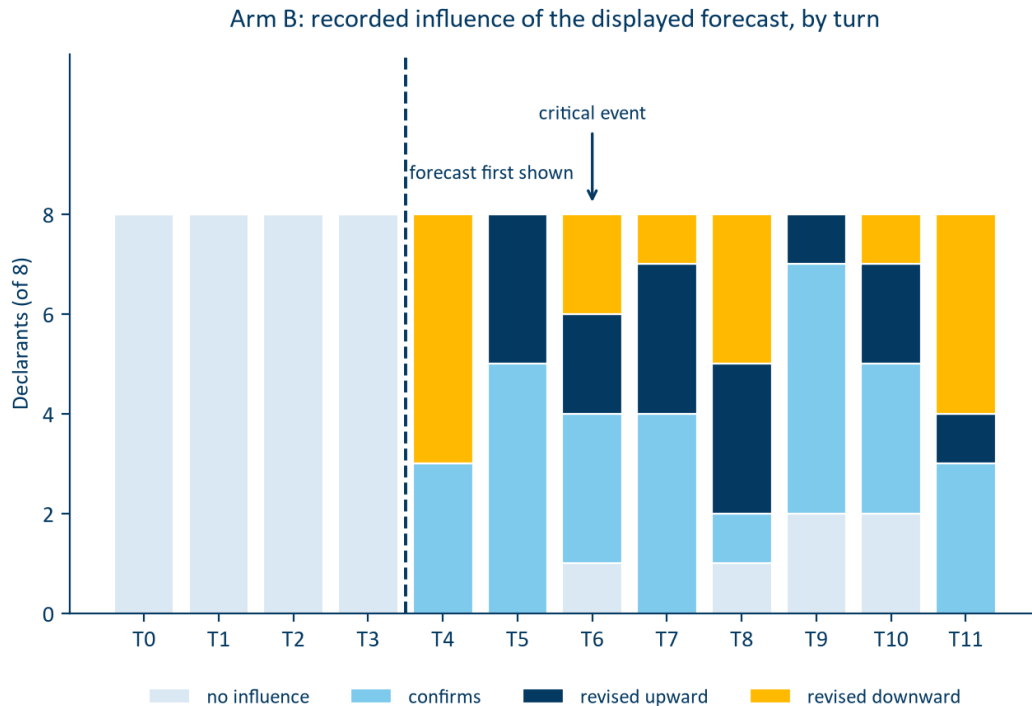


Figure 14: Influence of the displayed forecast, by turn

than being a permanent bias.

Two things make this worse than a curiosity. The forecast being displayed comes from the same layer whose confident predictions never materialise, and whose time block is blind to the very shock about to arrive. And it is displayed as a numeric probability with a confidence interval, a format the automation-bias literature identifies as increasing reliance (Goddard et al., 2012; Bansal et al., 2021). The system was presenting an unfounded reassurance in its most persuasive available form, immediately before the event it had failed to anticipate.

One qualification cuts against the strongest reading: the recorded field captures the effect on the declared level, not on the action taken, and the free-text notes show declarants still acting protectively. The effect is on the measurement, which is nonetheless the input the whole adequacy layer consumes.

7.4.5 The outcome definition moves the answer

A final measurement concerns how these results should be read at all. The identical one hundred and twenty forecasts were scored against three definitions of an adverse outcome, and the three disagree by more than any modelling choice made during the mission. Table 8 gives them at four weeks.

The first is the definition the running system applies, and it is the one the negative result above is stated against. The second counts any recorded event regardless of gravity and ignores milestones entirely; it is the easy target, and choosing it would report a system five times better calibrated without a line of code being touched. The third keeps the production definition of an adverse outcome but reads a missed milestone against the deadline the node originally committed to rather than against whatever deadline the milestone currently carries,

Table 8: The same forecasts under three definitions of truth, four-week horizon

Outcome definition	n	Base rate	AUC	Brier skill
Production, current deadline	120	0.050	0.485	−3.36
Naive, any recorded event	75	0.387	0.484	−0.64
Original commitment, censored points excluded	90	0.289	0.654	−0.55

and drops the forecasts whose observation window runs past the end of the campaign.

The model did not change between the three rows. Only the definition of truth did, and it moves the Brier skill by a factor of six and the area under the ROC curve from indistinguishable from chance to 0.654. The third row is the one the rest of this section uses, for a reason that is a matter of what a forecast is for rather than of statistics: a replanning is itself the admission of a slip, so a truth that lets a re-dating erase the miss scores the forecast against an outcome the forecast was supposed to warn about. A calibration figure quoted without its outcome definition therefore carries close to no information, which is a methodological result worth more to the programme than the score itself.

7.5 The Repair, and What It Measures

The diagnosis above named one component and specified a repair. That repair was then carried out and re-measured on the identical frozen campaign, which is what this section reports. The declarations are unchanged, the scenario is unchanged, and the same one hundred and twenty forecasts are recomputed by the modified model, so any difference belongs to the model alone.

7.5.1 The recommended repair was the wrong one

The obvious reading of the diagnosis is that the calibrated events should write a lead-time impact alongside their existing effects, and that is what the first version did. Measuring it showed it does not work, for a reason worth stating because it generalises. A completion date estimated as a full production cycle starting now scales the lead time by the fraction of work remaining, so anything routed into the lead time is scaled by that fraction too, and a shock therefore evaporates precisely on the milestones closest to their deadline, which are the ones that matter. On the node carrying the scenario's accident, the four-week probability fell from 0.846 to 0.000 on a milestone that was in fact missed. Lost production time is additive, not proportional: a four-week freeze costs four weeks whether the order is a tenth done or nine tenths done.

The completion model was rewritten accordingly, identically in the three independent estimators the system carries for the same quantity, the analytic time block, the Monte Carlo forecast rollout and the PERT lead-time simulator:

$$\text{completion} = t + (1 - \text{progress}) \times \text{lead time} + \text{lost time}, \quad (2)$$

a proportional cycle term and an additive lost-time term. Five families of capacity-halting event, machine failure, accident, strike weighted by the share of the workforce involved,

material shortage weighted by criticality and cyber incident, now accumulate their duration into the node's lost-time indicator. Loss of storage capacity deliberately does not, since it removes space rather than production hours.

The mechanical effect is the one the diagnosis asked for. Where the accumulated-delay field had been written a non-zero value zero times across the whole campaign, the node carrying the accident now accumulates 154.7 engine hours at its first shock and 371.2 over the campaign, carried in twenty-seven of its forty-one snapshots.

7.5.2 Two further defects the first measurement had hidden

Repairing the first defect exposed a second. With the time block finally responsive, the milestone probability turned out to be almost binary: on the frozen replay it returned exactly zero or exactly one in eighty-five percent of cases, leaving fourteen forecasts of one hundred and twenty anywhere in between. That is a switch, not a probability, and two of the declarants had said so in their free-text notes before it was measured. The cause is that the rollout drew the lead time at random but took the declared progress as exact. Declared progress is the least reliable input the system receives, and it is precisely the kind of quantity the whole mission exists to doubt: the tool measures the gap between declared and computed urgency because self-reported figures run optimistic, and then its own forecast was treating one of them as certain. Progress is now drawn per replicate, with a standard deviation widened node by node in proportion to that node's own measured hidden risk, so the system uses its own adequacy measurement to calibrate its own forecast.

The third defect was in the data rather than the code, and it is the one that would have invalidated any calibration attempted before it was found. Indicator coverage on the campaign database was 25.6%: twenty-four of the forty-two indicators were empty on all eight nodes, the carbon and cost blocks were inert everywhere, and only three of the six urgency blocks contributed anything at all. The lead time itself, the central quantity of the milestone channel, was present on five nodes out of eight, and the other three received a risk floor of zero through the documented fallback for a missing lead time, which is an absence of data presenting itself as a reassuring prediction. Filling the indicators from public figures of the real shortage raised coverage to 49.1% and made all six blocks active. One further bug became visible only after that: a per-node recovery capability, a twenty-six-week aerospace requalification time, was being read as production time already lost, which pinned that node's time block at 1.0 from the first turn. An empty database hides the errors in the model that consumes it.

7.5.3 What the re-measurement gives

Table 9 scores the recomputed forecasts against the original-commitment definition of Section 7.4.5, at all four horizons, on the same censored samples of 112, 104, 96 and 90 observations.

The structural repair improves both axes at every horizon, which is the clean result: the calibration deficit is roughly halved and the discrimination rises, and neither had to be traded for the other. The progress-uncertainty term is a trade rather than a gain. It improves calibration further at two weeks and beyond, worsens it slightly at one week, and lowers the area under the ROC curve; that last movement is not interpretable, since with twenty-six positive observations the ninety-five percent interval on the four-week figure runs from 0.21

Table 9: The frozen campaign rescored after repair

Horizon	Brier skill			AUC		
	before	structure	+ uncertainty	before	structure	+ uncertainty
1 week	−1.278	−0.646	−0.687	0.736	0.793	0.729
2 weeks	−0.690	−0.401	−0.367	0.636	0.671	0.634
3 weeks	−0.654	−0.360	−0.286	0.641	0.657	0.602
4 weeks	−0.553	−0.272	−0.216	0.654	0.721	0.584

to 0.87 and contains every value in its row. The term was therefore kept on the calibration evidence and on the argument that a declared quantity should not enter a probabilistic forecast as a certainty, not on the discrimination figure.

Three things should be said plainly against this. The Brier skill is still negative at every horizon, so the forecast is still worse calibrated than a forecaster that announces the base rate and stops. The milestone probability is still binary in seventy-two percent of cases, down from eighty-five, so the underlying over-confidence is reduced rather than removed. And a post-hoc calibration layer was trained, measured, and deliberately not connected: on the synthetic factory it improves the Brier skill by thirty-one percent, but on the campaign it improves calibration while dropping the four-week area under the ROC curve from 0.72 to 0.58, because it was trained on a target defined as an event striking the node at exactly one week while the product displays a missed milestone or an event cumulated over four. That is the trap of Section 7.4.5 met a second time from the other side: a layer calibrated against the wrong target degrades a system that works. Connecting it requires the training label to be rebuilt on the product’s target, which requires the synthetic factory’s snapshots to carry milestone truth, which they do not.

7.5.4 A third arm, and the honest reading of it

The repaired system was then run through a full nineteen-turn campaign in closed loop, in which the same eight declarants no longer only declared but also chose actions from a fixed catalogue, applied to the chain by the system and journalled: promote a backup arc, replan a milestone, expedite a shipment, raise capacity, revise a declaration, or do nothing. One hundred and fifty-two declarations were recorded with no unresolved consistency failure.

The headline figure flatters the tool and should not be used. Six of eleven milestones were missed in the frozen scenario and none was left unfinished in the closed loop, which sounds like six saved. Read against the original commitments it is not: five of eleven milestones were met on their original date in the frozen run, and five of eleven in the closed loop. The loop changed which milestone was met, not how many. One was genuinely saved, an electronics series delivered on its original date that had slipped in the frozen truth. Four late deliveries became replannings announced in advance instead of slips discovered at the deadline, which is worth something to a customer without being a save. And one milestone was replanned that would have been met, which is the measurable cost of over-reaction. With eleven milestones neither the benefit nor the cost is established; they are facts of one campaign, not results.

One further repair belongs here, because it touches the other half of the network layer.

The criticality ranking, which sweeps the network by knocking out each node in turn and measuring the damage at the final customer, collapses when many nodes saturate at once: every delta clips to zero and the ranking says nothing, in exactly the crisis where it would be worth reading. Nine of the nineteen campaign turns were in that state. The ranking now carries a second sort key computed in log-survival, $\ell(p) = -\ln(1 - p)$, on a regularised twin of the same propagation. Because ℓ is strictly increasing, the two keys order identically outside saturation, so nothing changes where the index already worked, and inside saturation the clip that flattens the first does not flatten the second. Measured on the same run, the number of turns on which the deep suppliers can be ordered at all rises from 10 to 17 of 19, and the expected bottleneck leads on 16 of those 17.

Two behavioural observations from that run do carry. The turn-four disarming of the first campaign reproduced exactly, eight declarants of eight. And the repair is visible in a declarant's own words at the turn the crisis event lands: where the first campaign's foundry manager wrote that the tool was holding a low position through a widely reported crisis and took no action, the repaired run moved the same node's four-week probability from 0.088 to 0.214 and the declarant replanned the milestone. The chain from a scripted shock to a displayed number to an action taken now closes, which it demonstrably did not before.

7.6 A Second Chain, Built to Break the Result

A result measured on one scenario is a property of that scenario until something tries to break it. The last phase of the mission therefore built a second supply chain whose design choices were made to be hostile to the first result rather than favourable to it, and ran the whole instrument through it again.

7.6.1 What was changed, and why each change is a threat

The second chain, AIRB, is an airframer's supply base: fifteen nodes over five tiers, from titanium and composite suppliers through structural, hydraulic and avionics subcontractors up to the final assembly line. Three properties separate it from the semiconductor campaign, and each removes a support the first result might have been resting on.

Figure 15 sets the two side by side. The scale change is the mildest of the three. The second is sharper: AIRB carries no exogenous event whatsoever, so a supplier degrades only because its own indicators drift, and any instrument that was implicitly reacting to a scripted shock has nothing left to react to. The third is sharper still. Milestone truth on AIRB is not written by the scenario author but computed from the indicators, through a health term that converts each week's lead time, equipment effectiveness and loss into a fraction of progress, accumulated until it reaches one:

$$\text{progress}_t = \min\left(\text{progress}_{t-1} + \frac{\text{health}_t}{d - s}, 1\right), \quad (3)$$

where s and d are the milestone's start and due turns. Whether a milestone is missed is therefore a consequence of the data rather than a decision, which removes the author from the answer. Twenty-two milestones are observable on the chain and five are missed, spread over three suppliers.

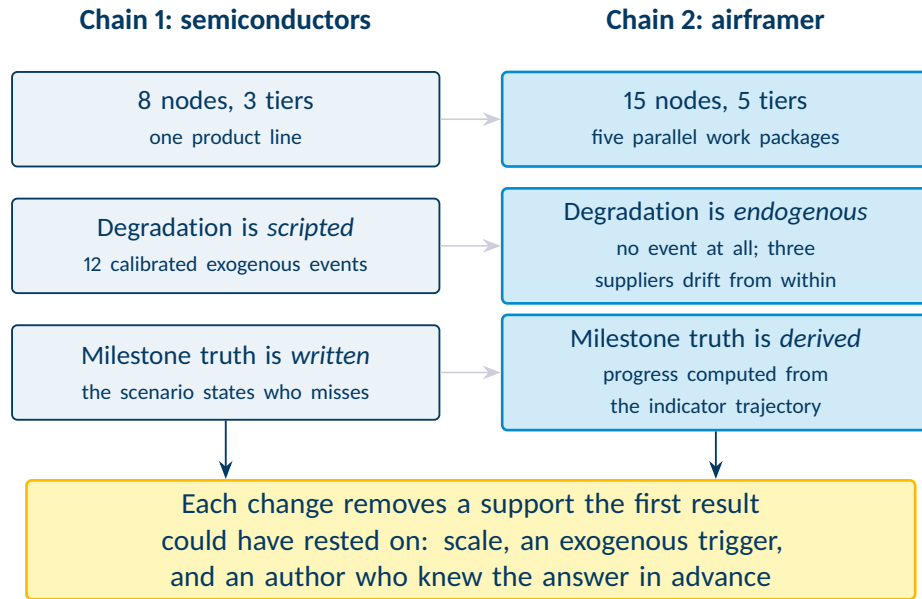


Figure 15: The two campaigns. The second chain was specified to be hostile to the first result, not to reproduce it.

7.6.2 The node-level result does not transfer, and that is the finding

The instrument ran nineteen turns with fifteen isolated declarants, producing 285 declarations. The adequacy signal, which on the first chain flagged three future failures with no false alarm, selects nothing at all here.

Table 10: Which ranking finds the three failing suppliers, second chain

Ranking	Precision@3	Turns	Reading
Adequacy score	0.00	19	Never selects a failing supplier
Hidden risk	0.00	19	Never selects a failing supplier
$[U_r - U_d]^+$			
Forecast probability	0.56	15	Selects nearly three times the base rate
Base rate	0.20	19	3 failing suppliers of 15

Table 10 states it without softening. Across all nineteen turns, neither the adequacy score nor the hidden-risk component ever places a future failure in its top three, against a base rate of 0.20 that random selection would have achieved. The forecast layer, on the same turns and the same truth, holds 0.56.

The reason is visible in the first turn's declarations and is worth more than the result it explains. Table 11 gives the four highest and the three lowest declared urgencies at turn 0.

The three suppliers that go on to miss declare the first, second and fourth highest urgencies in the chain, from the opening turn, before anything has visibly gone wrong. A quantity defined as the shortfall of declared against computed cannot open on a node whose declaration is already near its ceiling: their hidden risk is exactly zero for all nineteen turns, while the largest value the measure reaches anywhere on the campaign is 0.812, on nodes that hold.

Table 11: Declared urgency at the opening turn, second chain

Node	Declared	Computed	Hidden risk	Outcome
Titanium supplier	0.982	0.655	0.000	<i>misses</i>
Composite supplier	0.941	0.838	0.000	<i>misses</i>
Machining	0.914	0.811	0.000	holds
Harness maker	0.876	0.542	0.000	<i>misses</i>
Structures	0.535	0.990	0.456	holds
Systems integration	0.496	0.952	0.457	holds
Final assembly	0.400	1.000	0.600	holds

What makes this diagnostic rather than merely negative is the free text submitted alongside those numbers. In the same forms, the three suppliers write that they intend to delay informing their customer until they have a recovery plan to present with the bad news. They minimise in action and maximise in measurement, and the two are not in conflict: a supplier who has understood its own situation perfectly well, and is managing the disclosure of it, declares high and acts late.

The boundary this establishes. The adequacy gap detects a declarant who *misjudges* their own state. It cannot detect a declarant who judges it correctly and manages what they say about it. The first is a perception problem, and a questionnaire can reach it. The second is a disclosure problem, and no instrument that asks the supplier can reach it.

That distinction was not in the specification, was not in the state of the art, and would not have appeared on a chain where the failing nodes were also the under-declaring ones. It is the most transferable result of the mission, because it tells a future user of the instrument which of their two problems it is for.

7.6.3 Displaying the forecast, at five times the scale

The second chain was then replayed with the forecast displayed, on the same indicator pack and the same derived truth, so that display is the only difference between the two runs.

The first result is negative and simplifies the method for everything that follows. The forecast scores identically in both runs: a Brier skill of -1.601 against -1.592 at one week, an area under the ROC curve of 0.826 in both, and an unchanged node ranking. Displaying a forecast does not corrupt it, because the forecast is computed from indicators the declaration channel never touches. A treatment arm can therefore be scored as a control arm, which the design of this campaign had assumed to be impossible.

What the display changes is the declaration, and at fifteen declarants the pattern is far clearer than at eight. Figure 16 shows it turn by turn.

Three readings follow. First contact reproduces the eight-declarant result at nearly twice the scale: on the turn the forecast appears, eleven of fifteen report it confirms what they already thought, two revise downward, and none revises upward. One of them records the mechanism in its own words, writing that a work package stalled at zero progress for four consecutive turns should have raised its urgency further, but that the tool's reading of eleven

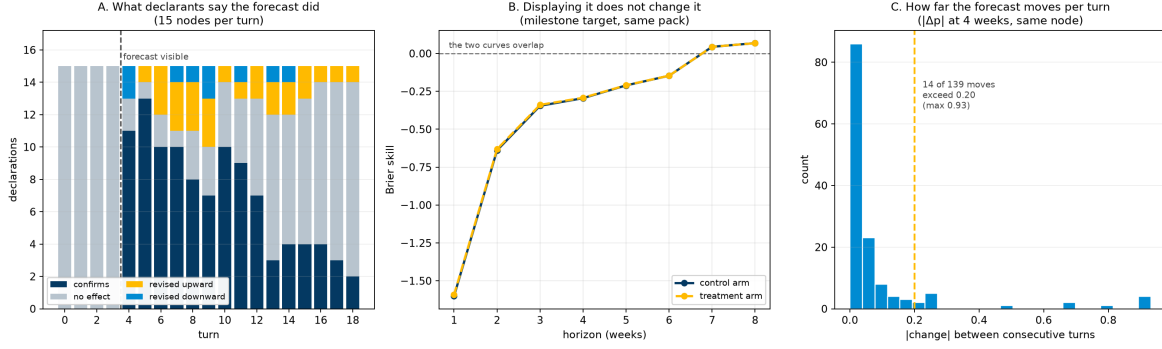


Figure 16: Second chain, run twice on one indicator pack. Left: what declarants report the forecast did to their judgement, by turn. Centre: the forecast scores identically whether or not it is shown, so the two curves overlap. Right: how far the four-week probability moves between consecutive turns on an unchanged node, which is what declarants cite when they stop reading it.

per cent tempered that concern.

Across all fifteen exposed turns, however, upward revisions outnumber downward ones 26 to 9. The one-sided suppression belongs to first contact; the persistent effect is not sedation but transfer. A declarant follows the number in whichever direction it points, which is a gain when the number is right and a loss when it is not.

The third reading is the one an engineer should act on. Reported influence falls from thirteen declarants of fifteen at first exposure to three at the final turn, and the share reporting no effect rises from two to twelve. The declarants give their reason in the notes, and it is measurable: between consecutive turns, on the same node with the terrain unchanged, the four-week probability moves by more than 0.20 on roughly one node in ten, with a largest single move of 0.93. An advisory number that jumps like that teaches its reader to stop looking, and disengagement is the rational response rather than negligence.

7.6.4 Does the forecast layer earn its complexity?

One question remained, and it is the one a reviewer asks last and an engineer should ask first. The forecast layer holds 0.56 precision where the adequacy gap holds 0.00, and it does so by running five hundred Monte Carlo trajectories per node per turn. Is that machinery doing the work, or is something much cheaper doing it?

Two observations raised the doubt. Enriching two dormant indicator blocks on this chain moved computed urgency on 174 of 285 node-turns and left the forecast identical to nine decimal places on 151 of 154 comparable points. Reading the simulation loop explained why, and Figure 17 draws it.

The consequence is testable, so it was tested rather than argued. A baseline was written to be as weak as possible: for a node whose next unfinished milestone falls due at turn D with fraction r of its work outstanding and nominal lead time L expressed in turns,

$$p(k) = \text{clip}\left(1 - \frac{D - t - k}{rL}, 0, 1\right). \quad (4)$$

No simulation, no fitted parameter, no calibration, and none of the six urgency blocks. The running service declines to forecast on 71 of the 225 node-turns, where no milestone is in

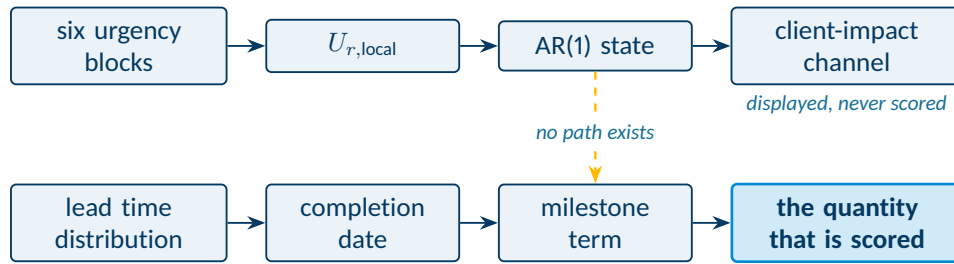


Figure 17: Inside the forecast rollout, the autoregressive state fitted to computed urgency feeds only the client-impact channel. The scored quantity depends on a lead-time draw and, on a chain with no exogenous event, on nothing else.

flight; the baseline is silenced on exactly those, so both are scored on an identical sample at every horizon.

Table 12: Five hundred Monte Carlo trajectories against a two-number ratio

Horizon	AUC		PR-AUC		Brier skill	
	simulation	baseline	simulation	baseline	simulation	baseline
1 week	0.826	0.908	0.153	0.396	−1.601	−4.851
2 weeks	0.781	0.848	0.212	0.261	−0.639	−4.289
3 weeks	0.757	0.791	0.252	0.258	−0.345	−3.949
4 weeks	0.697	0.731	0.257	0.223	−0.295	−3.990
5 weeks	0.651	0.676	0.275	0.221	−0.211	−3.798
6 weeks	0.616	0.624	0.301	0.237	−0.147	−3.486
7 weeks	0.667	0.583	0.438	0.239	+0.044	−3.180
8 weeks	0.633	0.544	0.460	0.228	+0.068	−2.961

Table 12 gives the comparison, with the better of each pair in italics. The baseline ranks the misses better than five hundred trajectories at six horizons of eight, and by 0.082 of AUC at one week, which is the horizon a coordinator acts on.

Since that baseline reads the outstanding fraction from the same progress variable the truth is derived from, it was re-run in a form that reads nothing at all: outstanding work taken as the calendar fraction a perfectly healthy milestone would show, using only the due date and the nominal lead time. Its AUC and PR-AUC are identical to three decimal places at every horizon. The discrimination is entirely the ratio of weeks remaining to weeks required, and no part of it comes from the indicators.

The same table says what the simulation layer does contribute, and it is not discrimination. Its Brier skill is −1.601 at one week against the baseline’s −4.851: the baseline produces an ordering and has no means of turning it into a probability, and converting an ordering into a scale is precisely what five hundred trajectories are for. It also holds its ranking at seven and eight weeks, at nearly double the PR-AUC, where a calendar ratio collapses below 0.60.

What this establishes, and what it does not. On this chain and this outcome, the forecast layer is a calibration wrapper around a lead-time ratio. That is a defensible thing for it to be, and at long horizons it is the better instrument. It is not what the interface presents, and it is not what the declarants of Section 7.6.3 believed they were deferring to. The

defect is nameable and cheap: the autoregressive state must enter the milestone term and not only the client-impact term. The repair would be measurable on these same frozen forecasts, and it is the first item of Section 8.3.

Two limits bound the comparison. It holds for a chain carrying no exogenous event, where the event term of the scored quantity is empty by construction; on the semiconductor chain that term is populated and the argument does not transfer unchanged. And five positive observations at one week make an AUC difference of 0.082 an order of magnitude rather than an established gap.

Running this test was optional. It was run because a layer that cannot beat a two-number ratio should not be shown to an operator as a probability, and the only way to know was to write the ratio and score it.

7.7 Problems Encountered and How They Were Resolved

Five difficulties shaped the mission, beyond the risks of Section 5.5.

The first model could not represent the problem. The initial formulation expressed urgency as a single time-driven curve. It could not represent a node on schedule but out of capacity, which is a common and dangerous state. The resolution was to abandon it in April in favour of multi-block aggregation. The cost was roughly three weeks; the alternative was building a validated implementation of an inadequate model.

Measuring the gap requires two independent signals, which is harder than it looks. The same difficulty recurred three times in different disguises. A first synthetic dry run derived the declared value from the computed one, so no gap could appear. The blind arm used declarants who read their briefings so carefully that their declaration reproduced the computed state. The treatment arm closed the loop deliberately by showing the model's output to the declarant. Each time the symptom looked like a modelling problem and each time the cause was circularity in the measurement. The resolution was to state the independence requirement as a first-class design constraint and to record, for every result, which of the two signals could have contaminated the other.

Literature retrieval had a long lead time of its own. Sources were obtained through ESTIA and Cranfield library access, and a paper was never available within the hour. Treated as a supply problem rather than an inconvenience, it was resolved the way the rest of the mission treats lead time: retrieval was front-loaded into the first three weeks against a planned list of what later phases would need, and every retained source was registered with its argument, method and limits so that the appraisal outlives the credentials used to obtain it.

A test bench has to be able to contradict its author. The last phase's difficulty was neither technical nor organisational. Each of the three tests of Sections 7.4 to 7.6.4 was designed while the outcome was still unknown, by someone with an obvious interest in one of the two answers. The resolution was procedural and worth carrying to any future R&D project: freeze the pass mark before the measurement exists, define the outcome in the same document as the threshold, and score any new instrument against the cheapest thing that could have

replaced it. The last of those is what produced the baseline comparison, and it is the test the system came closest to failing outright.

Compute was limited to a CPU. No GPU was available. This bounded what could be attempted and is the reason one comparison was run at reduced scale. The resolution was methodological rather than material: choosing approaches whose cost is dominated by the quantity of data rather than by model size, and accepting the reduced scale explicitly in the reporting instead of quietly omitting the comparison.

8 Conclusion and Personal Assessment

8.1 Results Against the Objectives

Table 13 assesses each objective of Section 4.3 against its indicator.

Table 13: Objectives against outcomes

Status		Evidence
O1	Met	Six-block aggregation with bidirectional propagation and an asymmetric adequacy score, each term exactly decomposable; aggregation choice justified against the compensatory alternative
O2	Met	SupplyScore delivered: 73 modules, 1831 automated test functions, multi-tenant persistence, every write audited, operational and reloadable
O3	Met	Propagation and criticality services operational, validated by dedicated performance benchmarks at the scale of a real programme; the criticality ranking's collapse under network-wide saturation was found on the campaign and repaired (Section 7.5)
O4	Answered, with a boundary	At node level the signal is right on the first chain: over the six turns before the crisis is public it flags three nodes, all three of which later miss a commitment, with no false alarm (Section 7.4.1). As a calibrated probability it first failed for an identified cause, was repaired and re-measured (Section 7.5). The second chain and the baseline comparison then bounded both results and named what each instrument is for (Sections 7.6 and 7.6.4)
O5	Met	Scientific report delivered; every reported figure regenerated from raw artefacts by script rather than transcribed

Quantitatively. The delivered system covers the specified scope and is measurably solid: 1831 automated test functions expanding to 2044 collected cases, an auditable write path, and performance validated at the intended scale. Two campaigns were run to completion across two supply chains, 23 nodes and 33 observable milestones, producing 510 recorded declarations.

The measurements themselves are numbers rather than impressions. On the first chain the adequacy signal flags three of four future failures with no false alarm over six consecutive turns. The forecast layer's Brier skill improves from -1.278 to -0.687 at one week and from -0.553 to -0.216 at four after repair, roughly halving the calibration deficit at every horizon while remaining negative. On the second chain the adequacy signal reaches precision 0.00 against a base rate of 0.20 while the forecast layer holds 0.56, and a one-line baseline outranks the simulation at six horizons of eight, by 0.082 of AUC at one week.

Qualitatively. The mission set out to determine whether the declared-versus-real gap can be measured well enough to act on. It can, at the level a coordinator actually works at, which is deciding where to look, and on a chain where the failing supplier is also the one misreading its own state. The three tests that followed established the boundaries of that claim more precisely than a favourable result ever could have.

What the programme can now say about the instrument. It detects misperception, not discretion. It is an early-warning instrument, and it closes by construction once a crisis is public. Its forecast layer contributes calibration and long-horizon ordering rather than short-horizon discrimination, for a reason located in one line of the model. Each of these is a statement about where the tool applies, and each came from a test designed to make the tool fail.

That is the outcome worth defending. A system that had scored well on one chain, on a signal driven by a block that could not see the crisis, would have been adopted and trusted. The route from here is not a new idea but more chains, and Section 8.3 says which and why.

8.2 Limitations

Four limits bound what this report supports, and three of them are limits of sample rather than of design. None calls the construction into question, and all three are answered by running the instrument on more chains rather than by changing it.

The declarants were agents rather than human consultants, so every statement about declaration behaviour applies to that population. The positive class is small on both chains, four failures of eight nodes on the first and three of fifteen on the second, which bounds every claim in both directions: it is why the precision of 1.00 in Section 7.4.1 is reported as an encouraging observation rather than a rate, and why the AUC movements in Section 7.5 are stated as uninterpretable rather than as gains. And the carbon block remains the least exercised of the six for want of node-level data, the measurement gap noted in Section 2.3; the indicator completion of Section 7.5 narrowed it without closing it.

The fourth is methodological and does not dissolve with more data. Every score depends on which of the three outcome definitions of Section 7.4.5 it is read against, and no number in this report should be quoted without one.

8.3 Perspectives

For ALTEN. The work leaves a system, two campaigns and an ordered list of what to do next. The list is short because each item was produced by a measurement rather than by an opinion.

Connect computed urgency to the quantity being forecast. The autoregressive state fitted to urgency currently reaches only the client-impact channel, which is why enriching two indicator blocks moved urgency on 174 node-turns without moving the forecast at all, and why a calendar ratio outranks the simulation at short horizons (Section 7.6.4). The change is small, and its effect is measurable on the frozen forecasts already recorded, so the cost of finding out is one afternoon.

Fill the indicators before calibrating anything. The campaign that produced the first negative result ran on a quarter of them with half the urgency blocks inert, and one real bug stayed invisible until the database was full. No amount of calibration compensates for a missing input.

Rebuild the post-hoc calibration layer's training label on the target the product actually displays. The layer as trained today improves calibration and degrades ranking, and is deliberately left disconnected; connecting it first requires the synthetic factory's snapshots to carry milestone truth.

Make the forecast state its own ignorance, and smooth the milestone alarm. A layer that cannot anticipate exogenous shocks should not present a low probability in a confident numeric format, and an alarm that moves from 0.05 to 1.00 in one turn is correct and useless as advance warning. The disengagement measured in Section 7.6.3 is the cost of not doing this.

Then run a campaign with human participants, which is the only run that can decide the behavioural hypotheses as they were written.

One indicator class is missing rather than empty, and several declarants named it independently: the forward booking horizon, meaning how far ahead a node's capacity is already sold. In the real semiconductor shortage that signal preceded the degradation of every flow indicator by months, and it is structurally invisible to the model as built. It is a general gap rather than a semiconductor one, since any capacity-constrained link holds that information.

Beyond the model, the adequacy layer is designed to be coupled to DISCO rather than to remain standalone, and the network representation work carried out in parallel on the same programme is its natural counterpart.

For the author. The subject continues in the research thesis produced for Cranfield University, which treats the same measurements at greater depth. The question I would most like to pursue is the one the display experiment opened: what an advisory layer owes the person reading it when it does not know.

8.4 Personal Assessment

8.4.1 What I learned

The hardest problem of these months was not mathematical. It was recognising, three separate times and in three different disguises, that the system was measuring itself. A first dry run derived the declared value from the computed one. A blind campaign used declarants who read their briefings so carefully that their declaration reproduced the computed state. And the forecast layer, built to doubt self-reported figures, was itself treating a self-reported progress figure as certain. Each time the symptom looked like a modelling defect and each time the cause was circularity. I would now design the independence of two signals before designing either of them.

The second lesson concerns what counts as a result. For a long stretch I treated a negative outcome as something to fix before reporting it. The most useful findings in this report are negative, and none would have been found by a project optimising for a positive one. The discipline that follows from it is cheap and I now apply it by default: before trusting any new component, write the cheapest thing that could replace it and score them against each other. That habit cost an afternoon and produced Section 7.6.4.

With hindsight I would change two things. I would have scored the forecast layer against recorded outcomes far earlier, since doing so took a few hundred lines and revealed the

defect immediately where months of implementation had not. And I would have written the outcome definition into the pre-registration alongside the thresholds, having since measured how much it moves the answer.

8.4.2 Link with the ESTIA engineering competency framework

Table 14 maps the mission onto the competencies of the ESTIA framework that it genuinely exercised. Appendix C gives the detailed evidence.

Table 14: Competencies exercised during the mission

	Competency	Evidence from this mission
CI1	Analysis, synthesis, creativity, systemic and interdisciplinary approach	Combining operations research, behavioural science, probabilistic modelling and software architecture into one measurement
CI3	Working in an international and intercultural context	Mission conducted and reported in English, under a double-degree programme, alongside an English-speaking colleague on the same research programme
CI5	Identifying and solving complex or research problems using external resources	The whole subject, and a state of the art built across published literature and the patent record when neither alone settled the question
CE3	Leading and controlling a project against objectives, deadlines and risks	A four-touchpoint weekly cadence, six phases, a documented mid-mission replanning, and a risk register of which three items materialised
CE4	Taking the socio-ecological transition into account	Carbon modelled as one of six first-class blocks rather than as a correction, and the point where environmental data stops being available at decision granularity identified and quantified
CE5	Researching, selecting, qualifying and diffusing information	A candidate dataset rejected on inspection before use; a registered appraisal of every retained source; the same work written three ways for three different audiences
CST1	Defining an engineering strategy in a poorly known context	Converging a five-workstream assignment onto one defensible contribution, and abandoning the first model when it proved unable to represent the problem
CST2	Modelling and simulating a system, capitalising the knowledge produced	The six-block model, its propagation, two campaign simulations, and a reproducible derivation of every reported figure
CST7	Building models, methods and tools to aid engineering, in an R&D perspective	The delivered application, the pre-registered protocol and the falsification tests built against it, all designed to be continued by the programme rather than to end with the placement

8.4.3 What this mission clarified professionally

I came into this placement expecting the interesting difficulty to be in the mathematics. It was not. The mathematics behaved. What failed was the join between a model and the world it was supposed to describe, and then the join between a model's output and the person meant to act on it. Both failures were visible only because the work was measured against something external rather than judged on its own internal consistency.

That is what I now understand the research and development engineer's job to be, as distinct from the design engineer's or the consultant's: not producing a system that works, but producing the evidence that says where it works and where it does not, in a form the next person can check. The three tests in Section 7 are worth more to ALTEN than the application they were run against, because the application is one programme's asset and the method transfers to any of them.

It is the direction I want to continue in, and specifically at the boundary where a model meets the person who has to act on it. That is where the failures are least obvious and most consequential, and this mission was a fairly direct demonstration of why.

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A Autorisation d'usage de l'IA par l'élève

À compléter et signer par le tuteur entreprise.

Formulaire

Nom de l'apprenant(e)	John HOARAU
Nom du tuteur	Sébastien PERTHUISOT
Entreprise	ALTEN Sud-Ouest, Direction de l'Innovation
Période concernée	du 13 avril 2026 au 16 octobre 2026

Rappel des risques pour l'entreprise en cas d'usage d'outils d'IA générative

L'utilisation d'outils d'IA par l'apprenti(e), sans encadrement ni précautions, peut exposer l'entreprise à plusieurs risques :

- divulgation involontaire de données confidentielles ou stratégiques lors de la saisie dans des outils en ligne ;
- perte de maîtrise sur la propriété intellectuelle des contenus générés (textes, codes, visuels) pouvant être réutilisés par la plateforme ;
- dépendance à des outils tiers dont les résultats peuvent être imprécis, biaisés ou juridiquement discutables ;
- non-conformité au RGPD si des données à caractère personnel sont traitées par des services non conformes.

Autorisation du tuteur

☐ J'autorise l'élève à utiliser des outils d'intelligence artificielle générative, en pleine connaissance des risques mentionnés ci-dessus.

☐ Je n'autorise pas l'usage d'outils d'IA générative pendant la période en entreprise.

Commentaires ou restrictions spécifiques éventuelles :

Fait à _____ le _____

Signature du tuteur / maître de stage :

Declared use of AI tools in this mission

For transparency, and independently of the authorisation above, the use made of generative AI during this placement is declared here in full.

Large language models were used in two distinct capacities, which should not be confused.

First, as an *object of study*. The eight declarants of the experimental campaign reported in Section 7.3 are language-model agents. This is an experimental design choice, stated as such throughout the report, and its consequences for the validity of the results are discussed rather than glossed over.

Second, as a *writing aid*, for language correction and document formatting. Every scientific claim, every numerical result and every interpretation in this report is my own, and I am able to justify and explain each of them. Every figure was produced either by the project's own code or by hand. No confidential ALTEN data was submitted to any external service in the course of that assistance.

B Detailed Planning

This appendix expands the schedule of Section 5.4 into the weekly rhythm that produced it, and records the two planning revisions made during the mission.

Weekly cadence

Table 15 lists the standing meetings that structured every week of the mission.

Table 15: Standing weekly meetings

Day	Meeting	With	Purpose
Monday	Planning, Green Supply Chain activity	É. Lardy	Set the objective for the week
Tuesday	Scrum	É. Lardy	Progress, blockages
Wednesday	Scrum	É. Lardy	Progress, blockages
Wednesday	Supply chain regular point	S. Perthuisot	Scientific direction and arbitra- tion
Thursday	Scrum	É. Lardy	Progress, blockages
Friday	Demonstration	É. Lardy, S. Perthuisot	Show working output, not de- scribe it

Six standing contact points a week is a demanding cadence for a research placement. Its effect was to bound how long any misconception could survive: the Friday demonstration in particular required something to be shown rather than reported, which repeatedly exposed gaps between what was believed to work and what did.

Planning revisions

June: validation over extension. The original plan allocated the second half of the mission to widening the model's scope. It was reallocated to validating what already existed. The argument accepted at the regular point was that extending an unvalidated model compounds risk rather than reducing it. This revision is why the mission produced a measured result rather than a larger unmeasured system.

July: substitution of the campaign population. The campaign was designed for eight human consultants across nine working days. When that could not be secured within the mission window, the protocol was executed with isolated agents instead. The record format was deliberately identical, so the substitution changed the population without changing the instrument. Every behavioural conclusion was weakened accordingly in the reporting rather than being stated at the strength the original design would have supported.

Comparison of planned against actual

Table 16 compares the time allowed for each phase with the time it actually took.

Table 16: Planned against actual, by phase

Phase	Planned	Actual	Comment
Framing	3 weeks	3 weeks	On plan
First pipeline	6 weeks	5 weeks	Stopped early once judged inadequate
Redesign and build	4 weeks	3 weeks	Interfaces already established by the abandoned pipeline
Experimental design	2 weeks	3 weeks	Freezing the protocol properly took longer than allowed
Campaign and analysis	3 weeks	3 weeks	On plan
Falsification	<i>unplanned</i>	3 weeks	Added after the diagnosis: repair, second chain, baseline comparison
Capitalisation and reporting	4 weeks	in progress	Runs to the end of the mission

The overall schedule held, and the three weeks the falsification phase consumed were released by the two phases that finished early. The one systematic underestimate was the effort required to freeze an experimental protocol to a standard where the hypotheses could not later be adjusted, which took half again the time allowed for it and was worth every day.

What the remaining two months carry

The mission runs to 16 October and this report is written from 20 August. The remaining time carries no new construction. It is allocated to running the instrument against further supply chains, which is the single action that moves every sample-bound figure in Section 8.2, and to the reporting and defence obligations that close the placement. Any rate quoted in this report may therefore move; the constructions and the diagnoses will not.

C Detailed Competency Evidence

This appendix gives the evidence behind Table 14, competency by competency, so that each claim can be checked against something in the record rather than taken on assertion.

CI1 – Analysis, synthesis, curiosity, creativity and innovation, approaching a problem systematically and across disciplines. The measurement built during this mission required combining four fields that do not normally cite one another: deadline-driven value from operations research, the mere-urgency effect from behavioural science, probabilistic aggregation, and the software architecture needed to make the result auditable. The contribution claimed is not any one of them but the combination, which Section 6.8 shows to be absent from the reviewed literature.

CI3 – Working in an international and intercultural context. The mission was conducted, documented and defended in English, within a double-degree programme with a British university. A colleague on the same research programme worked in English on the adjacent representation problem, and the weekly demonstration was the point at which the two strands had to be made intelligible to each other.

CI5 – Identifying and solving complex or research problems, mobilising scientific and technical knowledge and external resources. The subject was a research problem with no established answer, and settling that it had none required two distinct bodies of evidence: the published literature, which says how to compute risk and how to elicit a judgment, and the patent record, which says what industry has protected. Neither alone would have closed the question, since a working technique can be commercialised without publication and a published one can go unclaimed (Section 6.6).

CE3 – Leading, steering and controlling a project to reach its objectives, in cost, deadline, performance and risk. Seven phases with dated milestones; a weekly cadence of six standing meetings; a documented mid-mission replanning that traded scope for validation; and a risk register of four items of which three materialised and were handled. An unplanned falsification phase was absorbed without moving the end date, using the margin released by two phases that finished early (Appendix B). Cost is the one dimension not exercised, since no budget was managed.

CE4 – Taking the socio-ecological transition into account. Carbon is one of the six blocks of the model, ranked alongside cost and delay rather than applied as a correction, and the programme itself falls within the share of ALTEN's R&D devoted to environmental innovation (Section 2.3). The honest qualification, stated in Section 8.2, is that this block turned out to be the least exercised of the six for want of data.

CE5 – Researching, selecting, qualifying and diffusing information, adapting it to its context. The qualifying step is the one worth evidencing: a candidate dataset was inspected before use and rejected on three measurable signatures of synthetic data, and the node it would have fed was driven by calibrated events instead. On diffusion, the same work was written three ways for three audiences: a scientific report for the company, a research thesis for the partner university, and this mission report.

CST1 – Defining an engineering strategy by choosing methods and tools suited to a known or poorly known context. The assignment as signed covered five workstreams. Converging it onto one defensible contribution, and recording the boundaries explicitly rather than leaving them implicit, was the first engineering decision of the mission. The second was abandoning the initial urgency formulation once it proved unable to represent a node on schedule but out of capacity.

CST2 – Modelling and simulating a system, and capitalising the information and knowledge produced. The six-block model with its bidirectional propagation; two campaigns, one replaying a documented historical crisis and one built with milestone truth derived from the indicators rather than scripted (Equation 3); and a derivation pipeline that regenerates every reported figure from raw artefacts so that no number in any of the three deliverables is transcribed by hand.

CST7 – Designing and building models, methods and tools to aid engineering and improve business processes, including in a research and development perspective. The delivered application and the experimental protocol were both built to be continued rather than to end with the placement: a documented architecture, an automated test suite, a frozen and hashed protocol, and a written record of the next steps in order.

The research contribution proper is the test bench rather than the tool. Three instruments were built to attack the system's own claims: a scoring against recorded outcomes, a second supply chain specified to be hostile to the first result, and a minimal baseline that asks whether the most expensive component earns its cost. All three transfer to any model this programme builds next, which is why they are documented as a method in the scientific report rather than as an episode of this mission.